



US012415125B2

(12) **United States Patent**
Kim et al.

(10) **Patent No.:** **US 12,415,125 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **METHOD FOR PROVIDING WORKOUT DATA USING A PLURALITY OF ELECTRONIC DEVICES AND ELECTRONIC DEVICES THEREFOR**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 286 days.

(21) Appl. No.: **17/893,644**

(22) Filed: **Aug. 23, 2022**

(65) **Prior Publication Data**

US 2022/0401815 A1 Dec. 22, 2022

Related U.S. Application Data

(63) Continuation of application No.
PCT/KR2022/007771, filed on May 31, 2022.

(30) **Foreign Application Priority Data**

May 31, 2021 (KR) 10-2021-0070325

(51) **Int. Cl.**

A61B 5/11 (2006.01)

A61B 5/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **A63B 71/0622** (2013.01); **A61B 5/1114**
(2013.01); **A61B 5/4561** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC A63B 2220/803; A61B 5/4561

See application file for complete search history.

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Primary Examiner — John E Breene

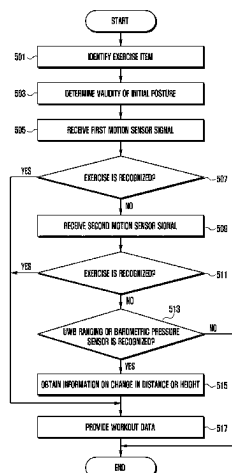
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(57) **ABSTRACT**

An electronic device is provided. The electronic device includes a display, a communication circuit, and at least one processor configured to recognize a user's initial posture based on at least one of motion sensor signals of a first external electronic device and a second external electronic device received through the communication circuit, control the first external electronic device to receive a first motion sensor signal, obtain the user's workout data based on the first motion sensor signal when the initial posture satisfies a designated condition, control the second external electronic device to receive a second motion sensor signal when the user's movement is not recognized based on the first motion sensor signal and obtain the user's workout data based on the second motion sensor signal, and provide the workout data through at least one of the display or the first external electronic device.

20 Claims, 15 Drawing Sheets



- (51) **Int. Cl.**
A63B 71/06 (2006.01)
G01P 13/00 (2006.01)
G06F 3/14 (2006.01)
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- (52) **U.S. Cl.**
CPC **G01P 13/00** (2013.01); **G06F 3/14**
(2013.01); **A63B 2071/0663** (2013.01); **A63B**
2220/20 (2013.01); **A63B 2220/74** (2013.01);
A63B 2220/803 (2013.01)

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FIG. 1

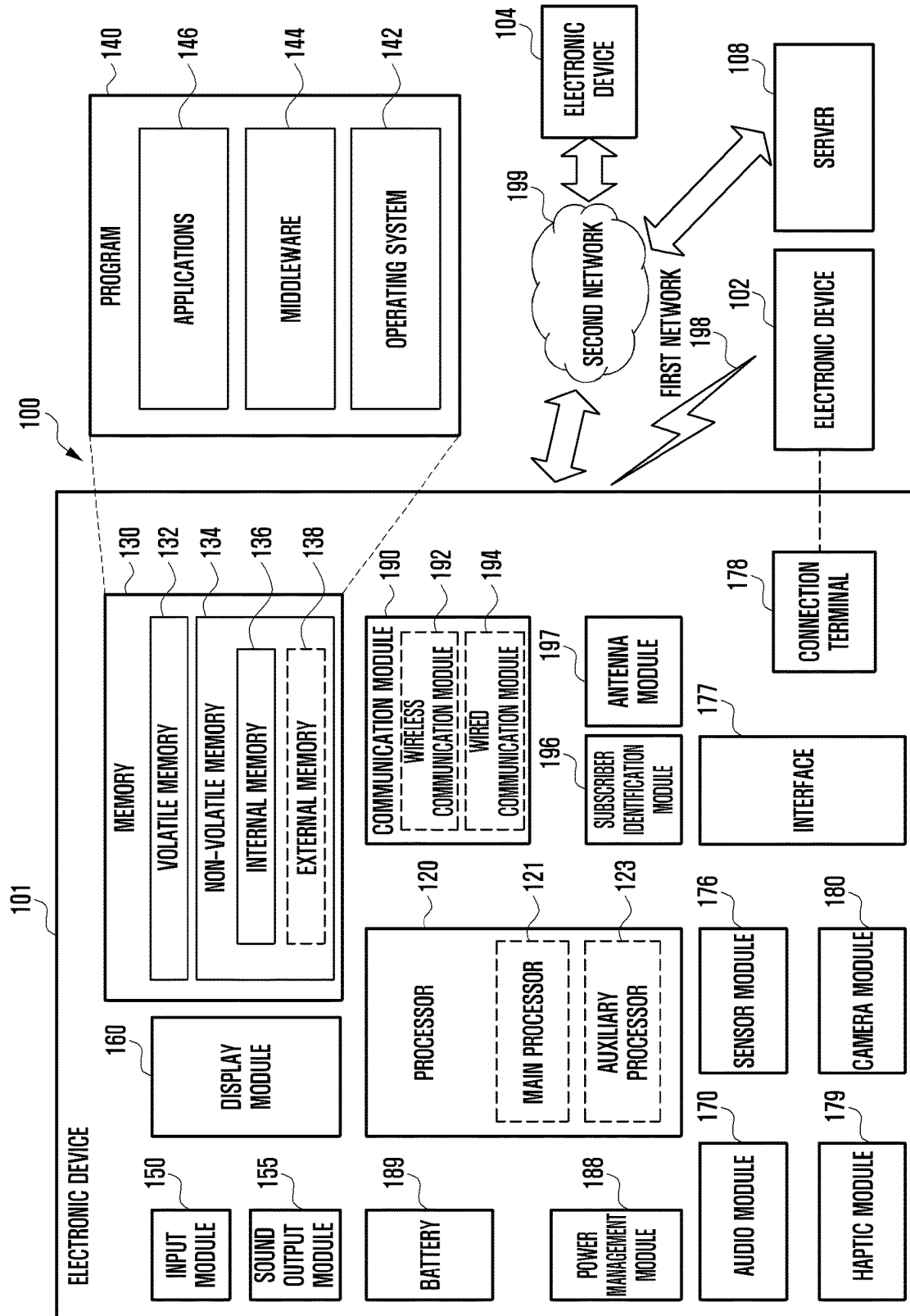


FIG. 2

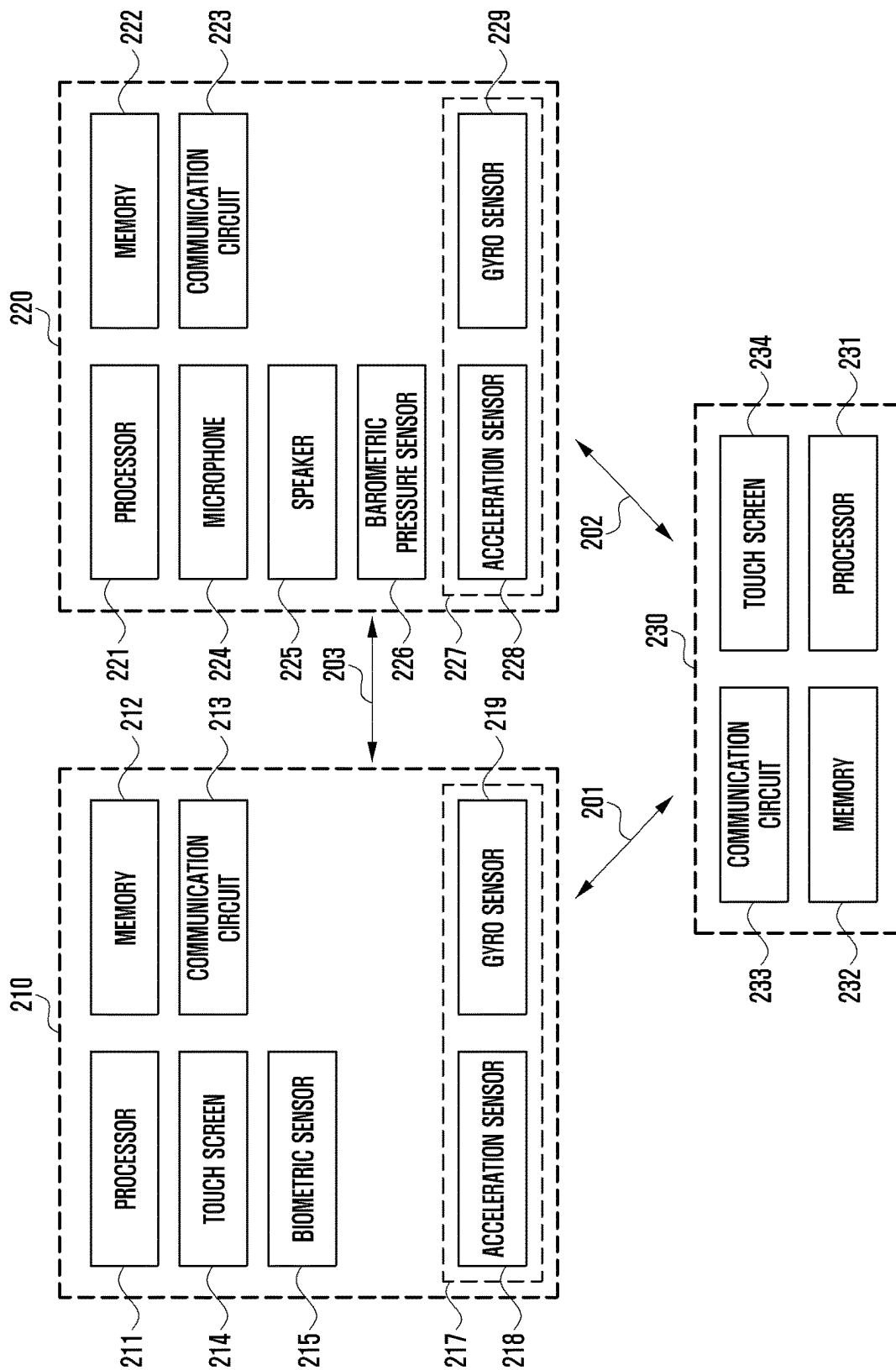


FIG. 3

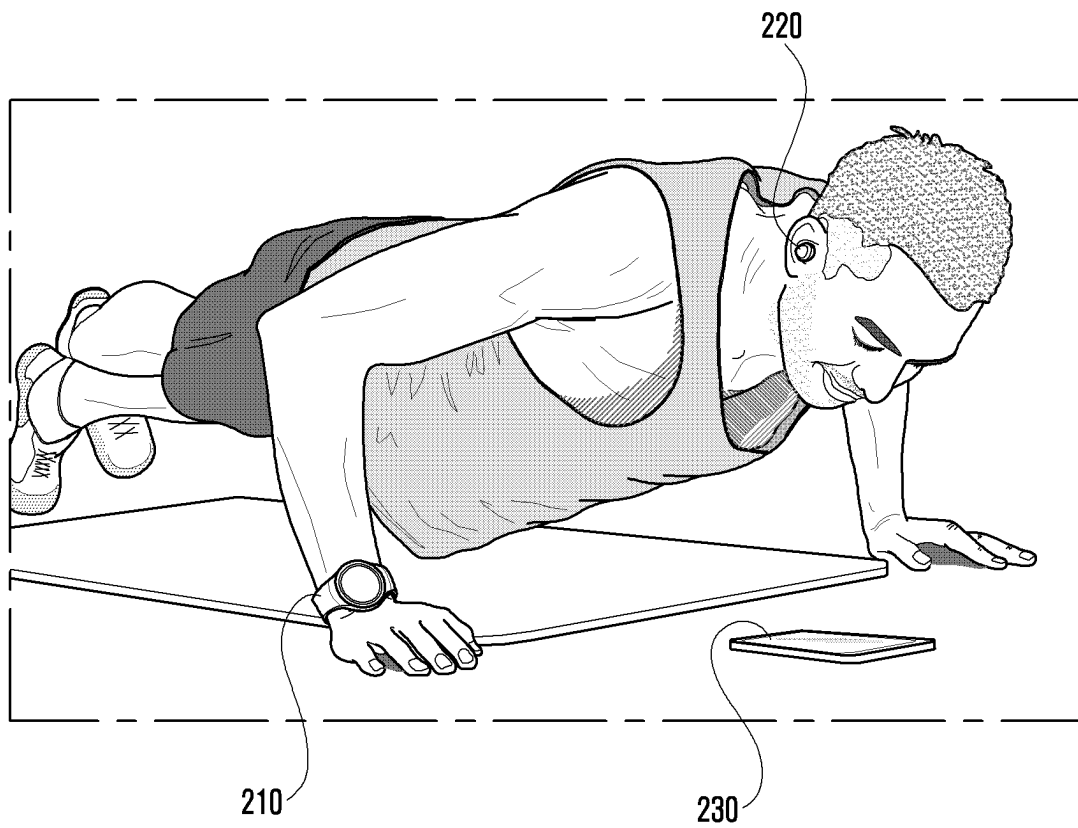


FIG. 4

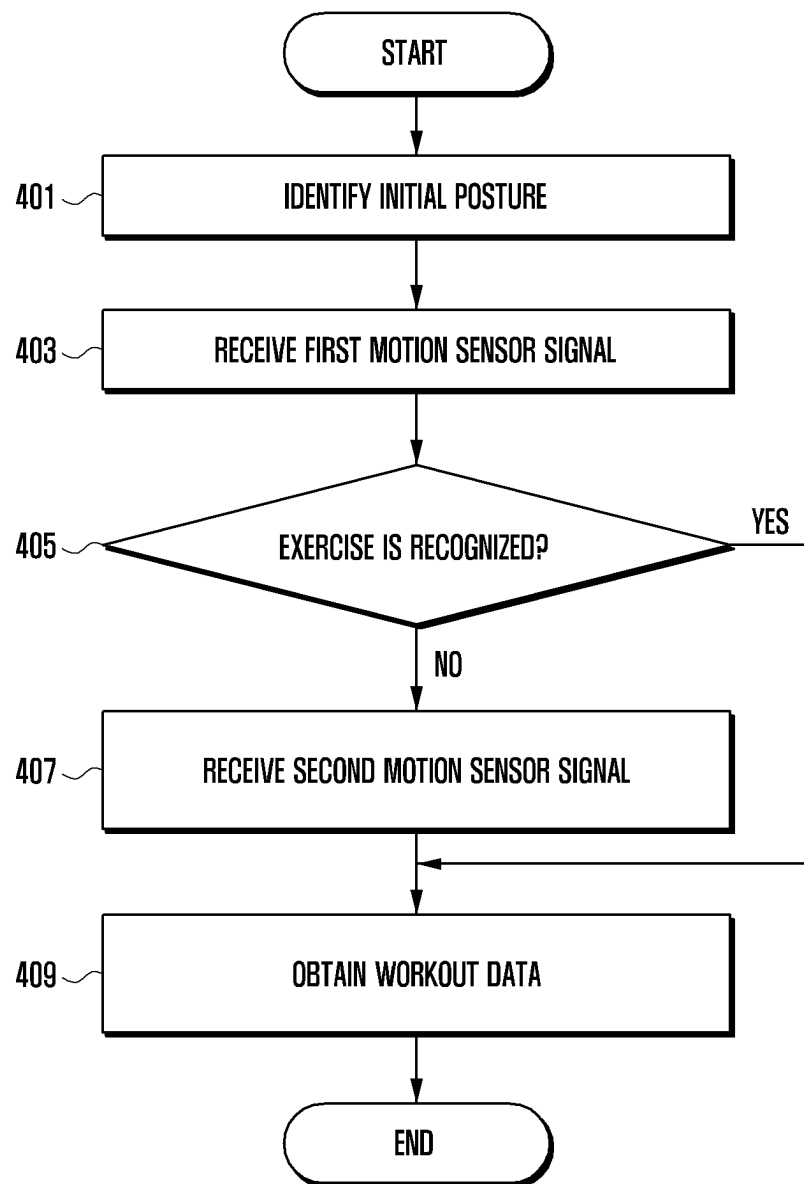


FIG. 5

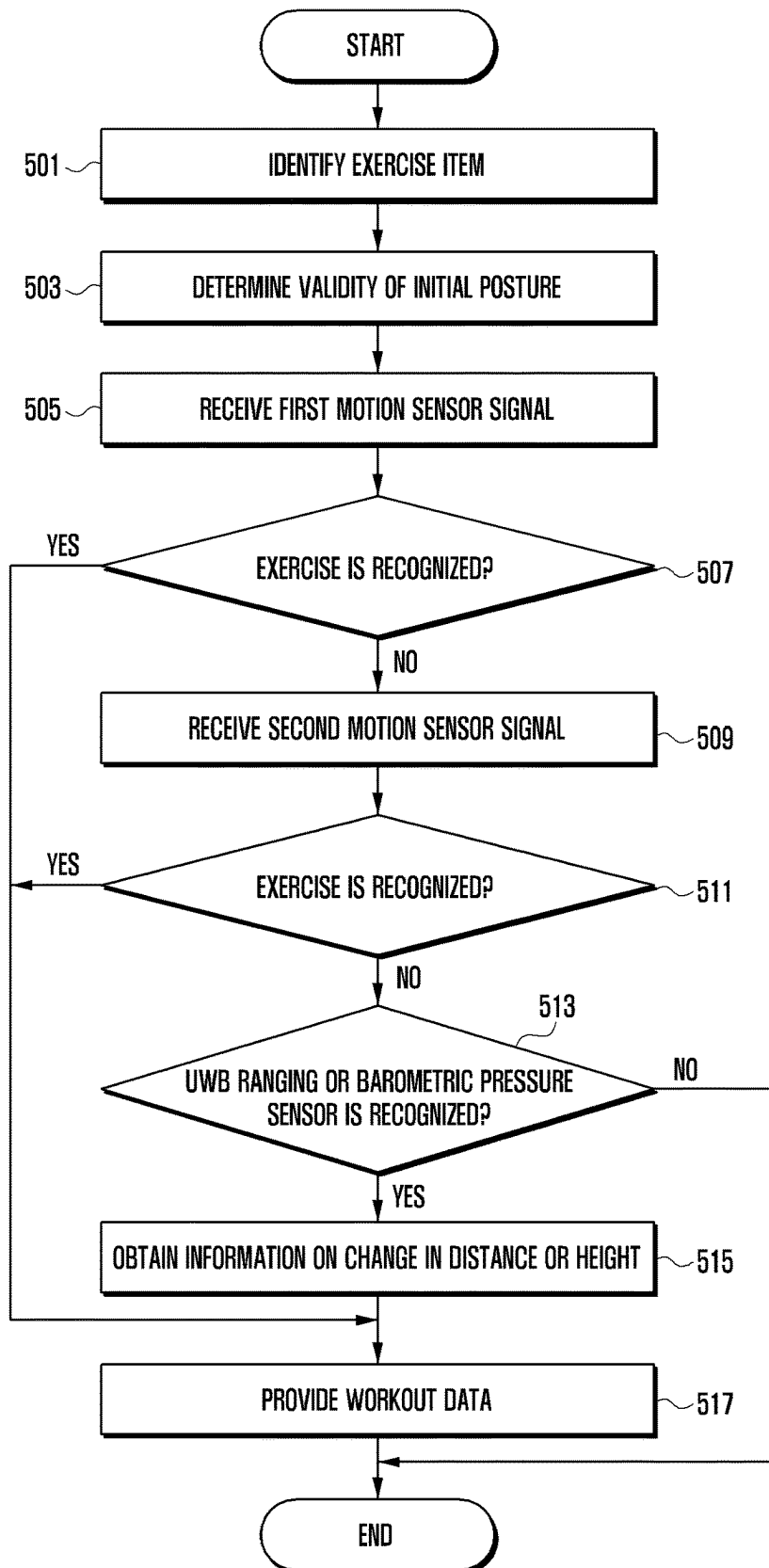


FIG. 6

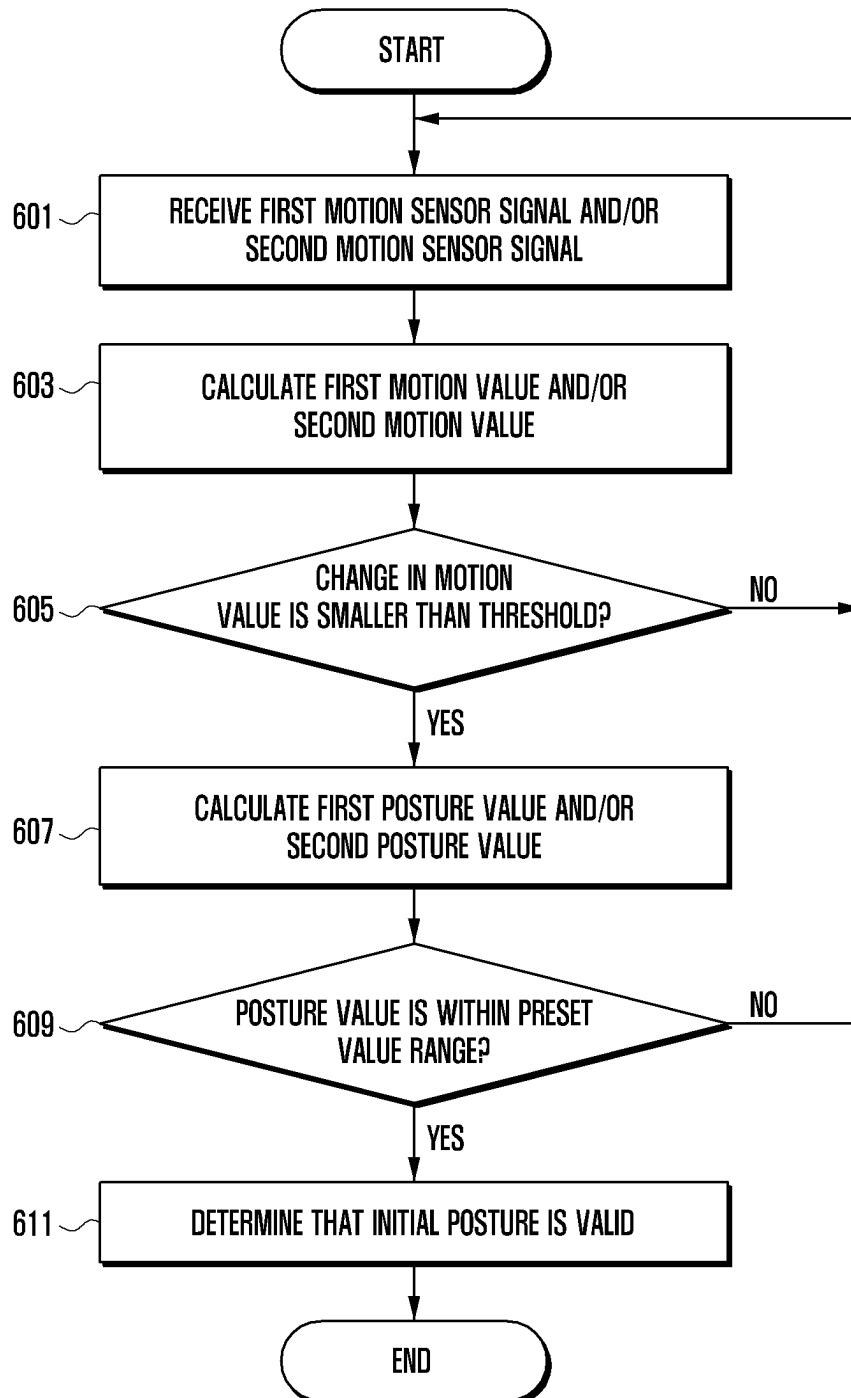


FIG. 7

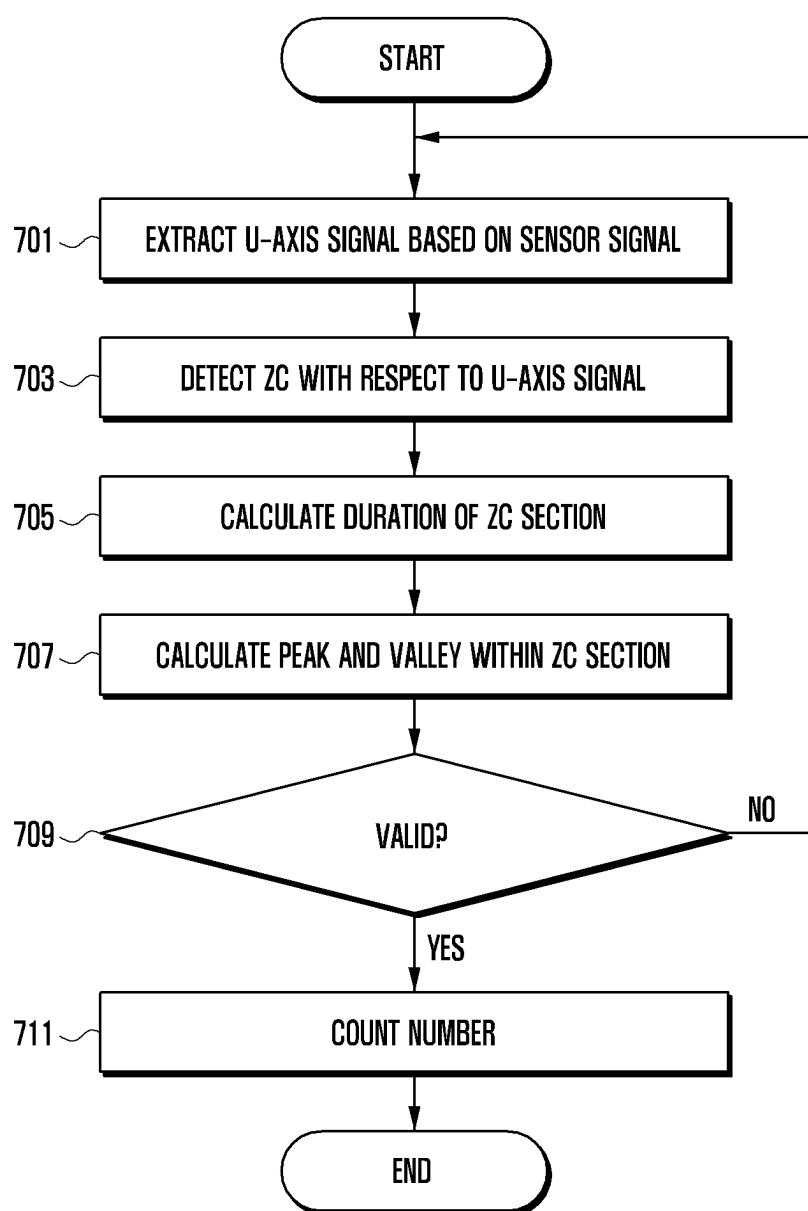


FIG. 8

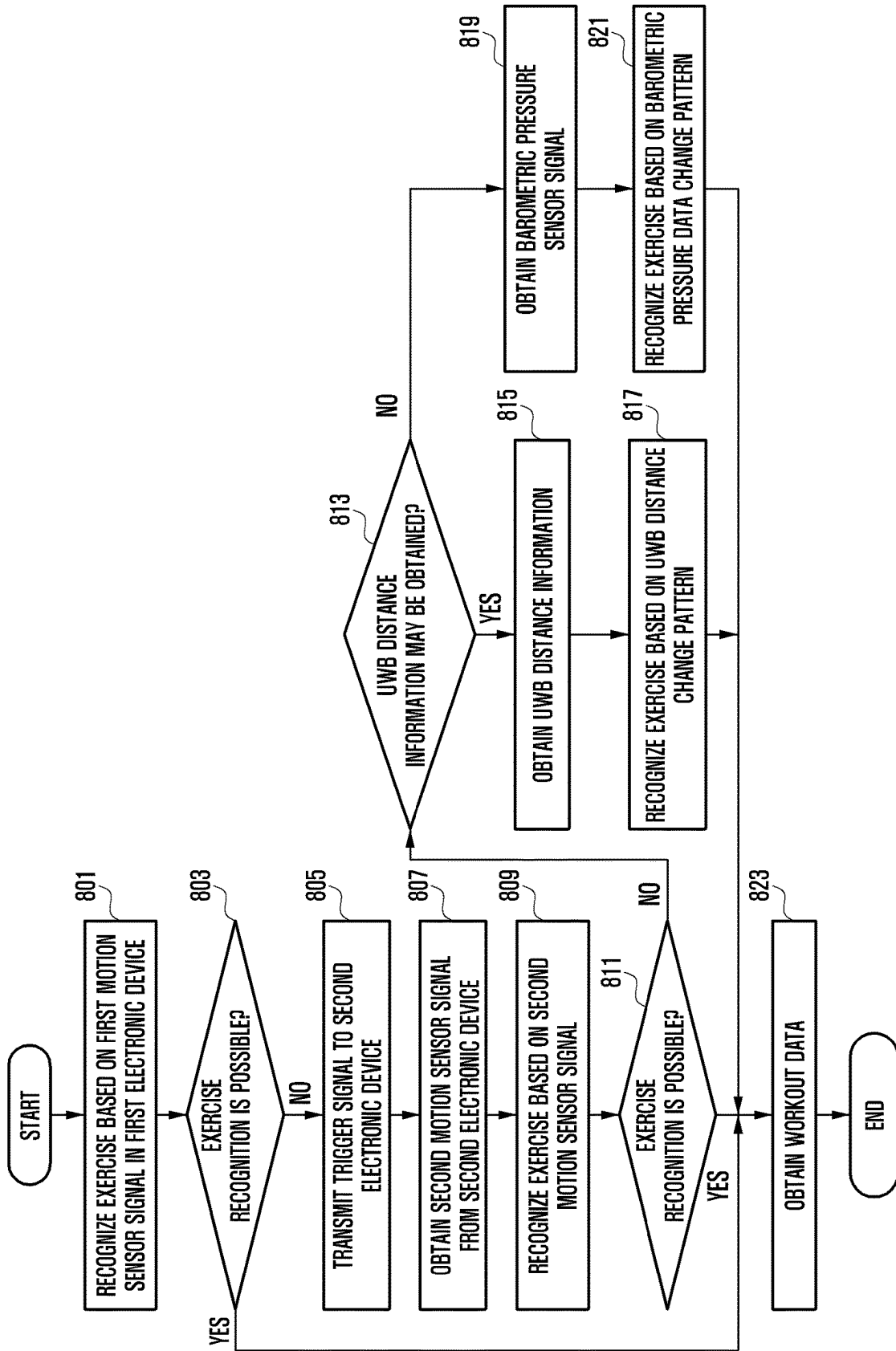


FIG. 9A

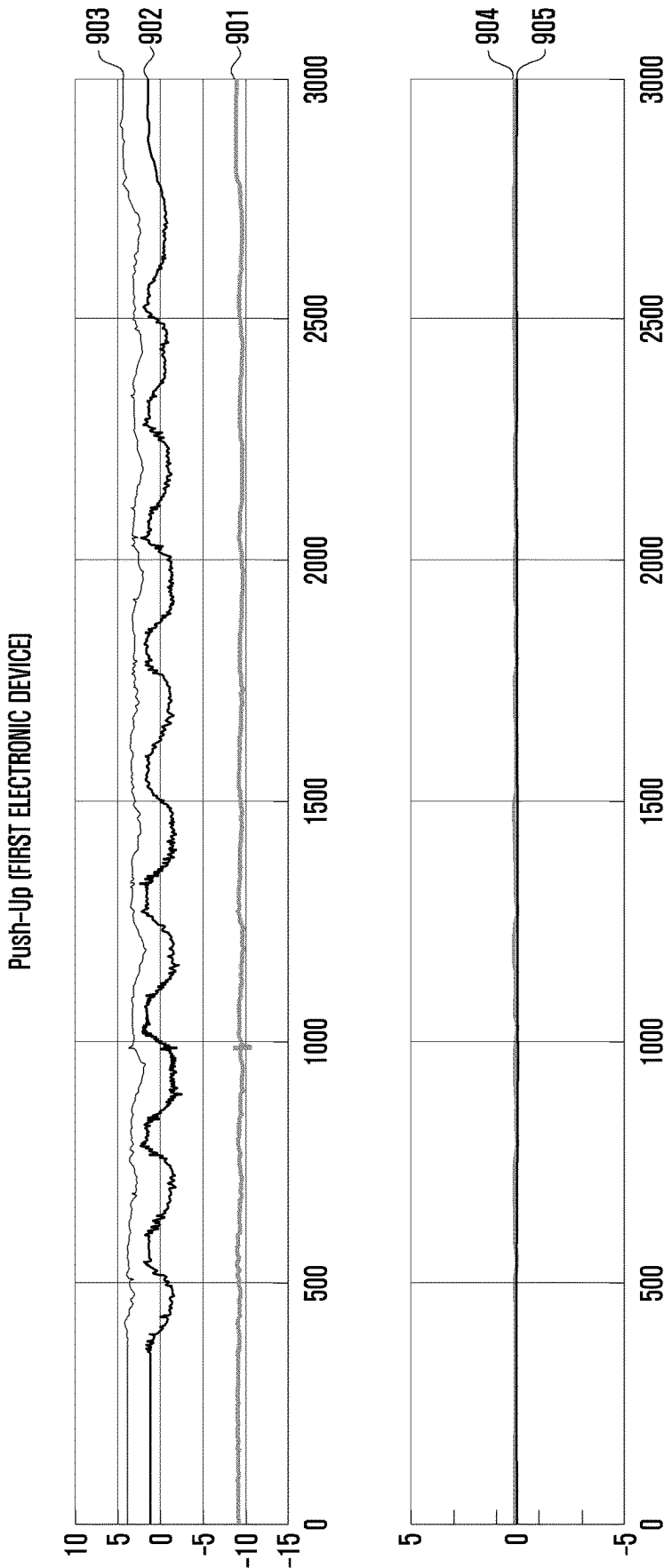


FIG. 9B

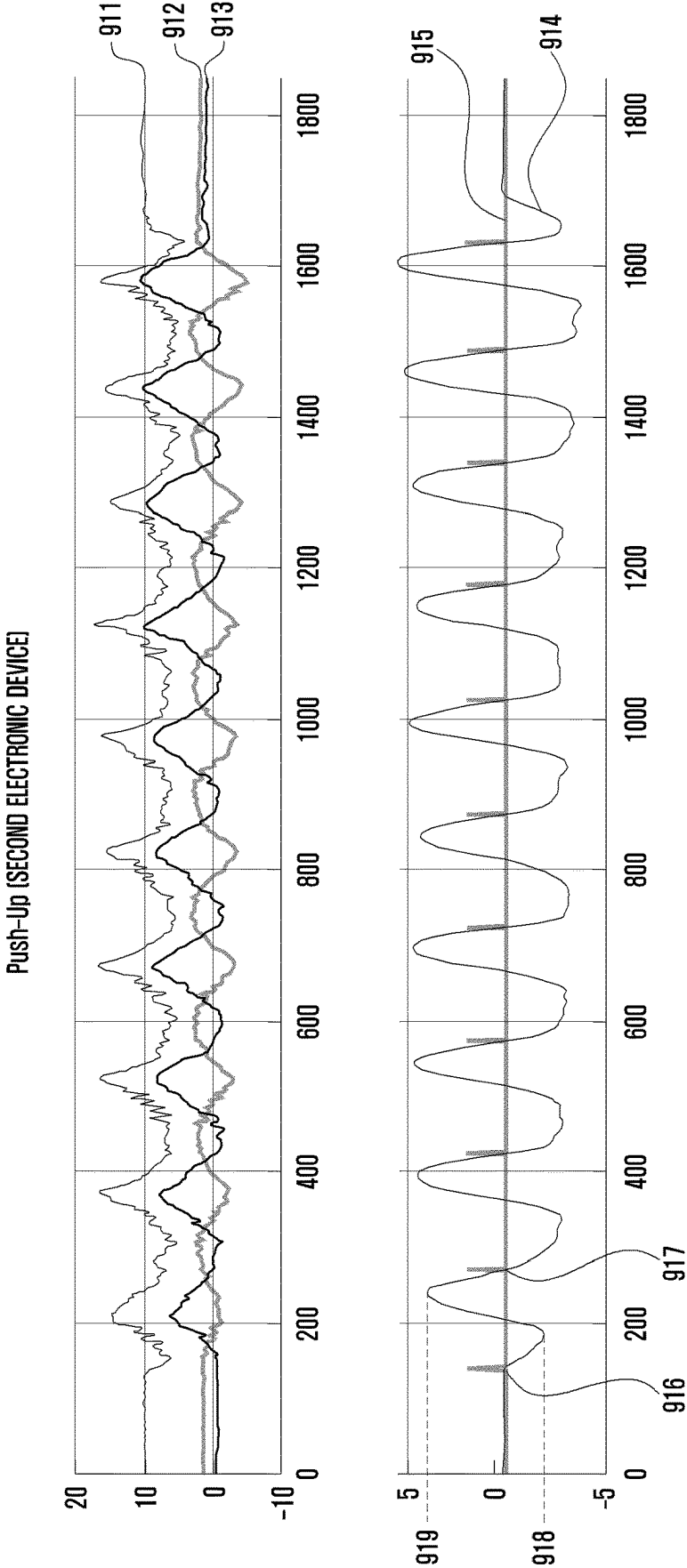


FIG. 9C

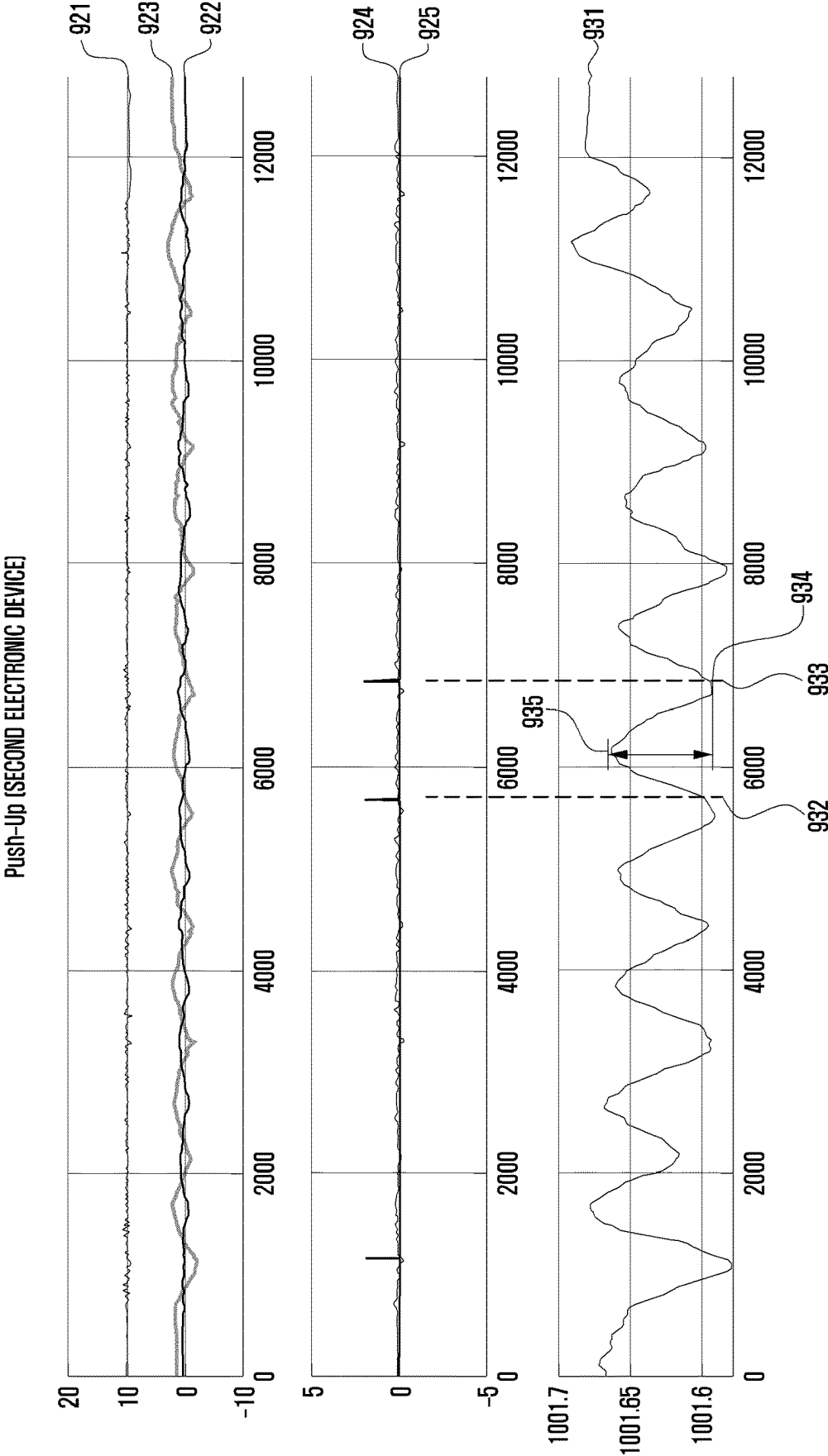
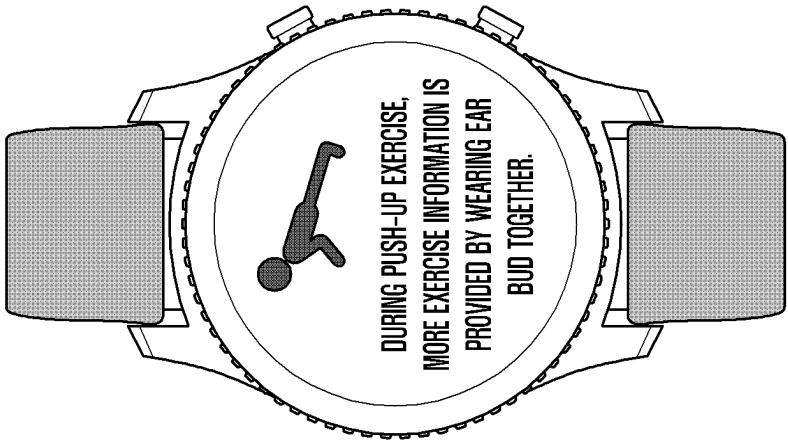
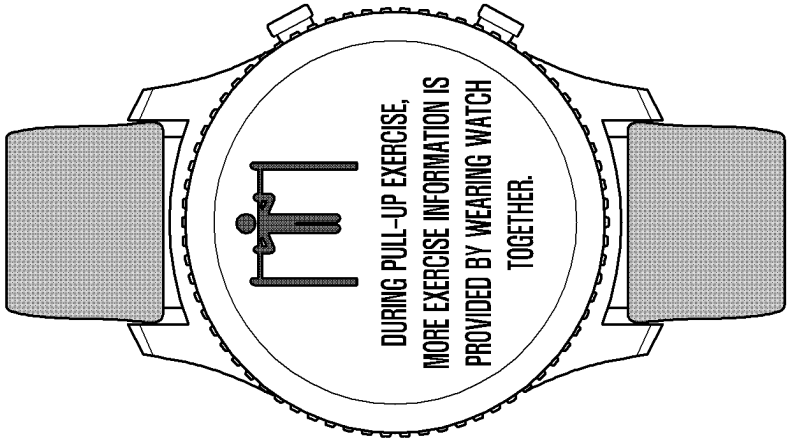


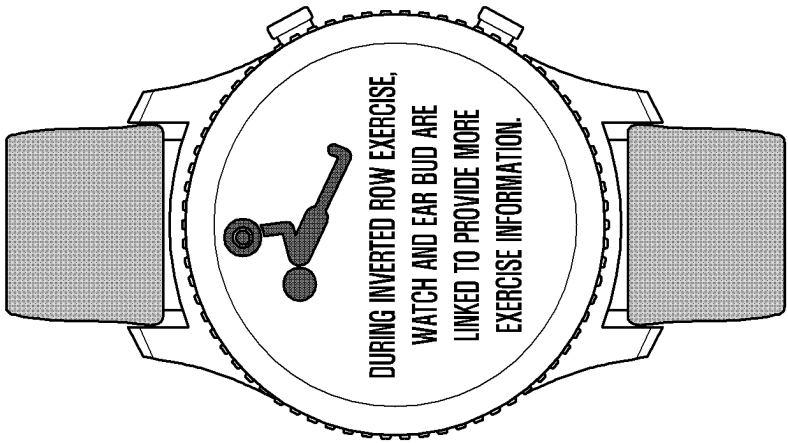
FIG. 10



(a)

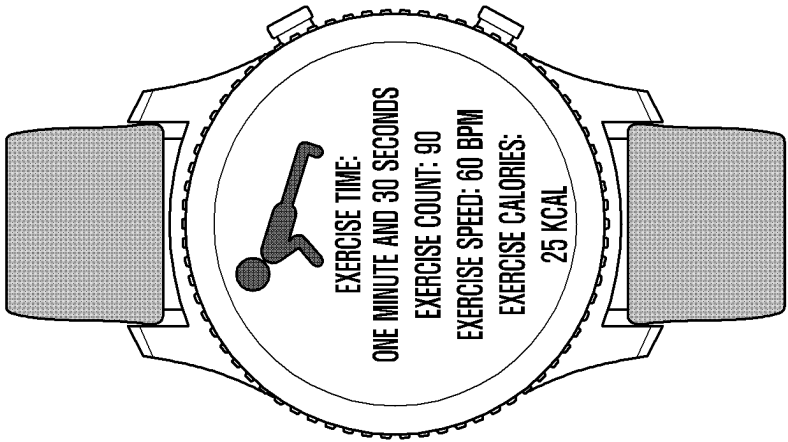


(b)

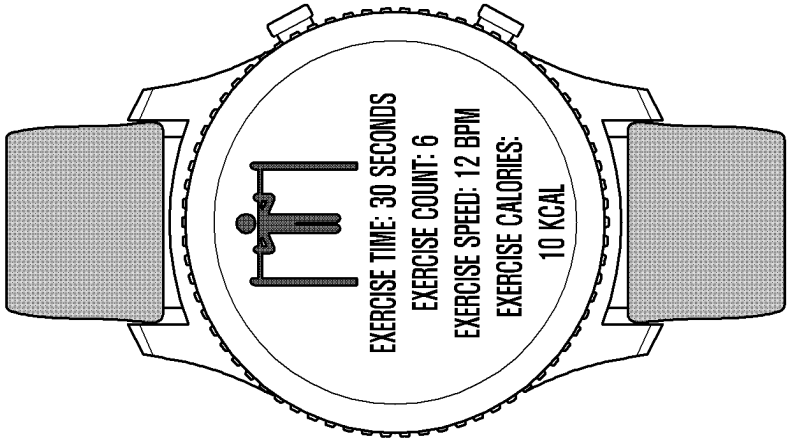


(c)

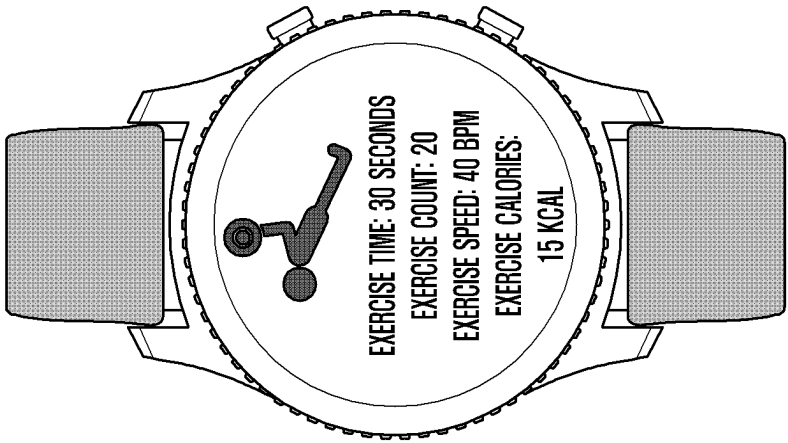
FIG. 11



(a)



(b)



(c)

FIG. 12

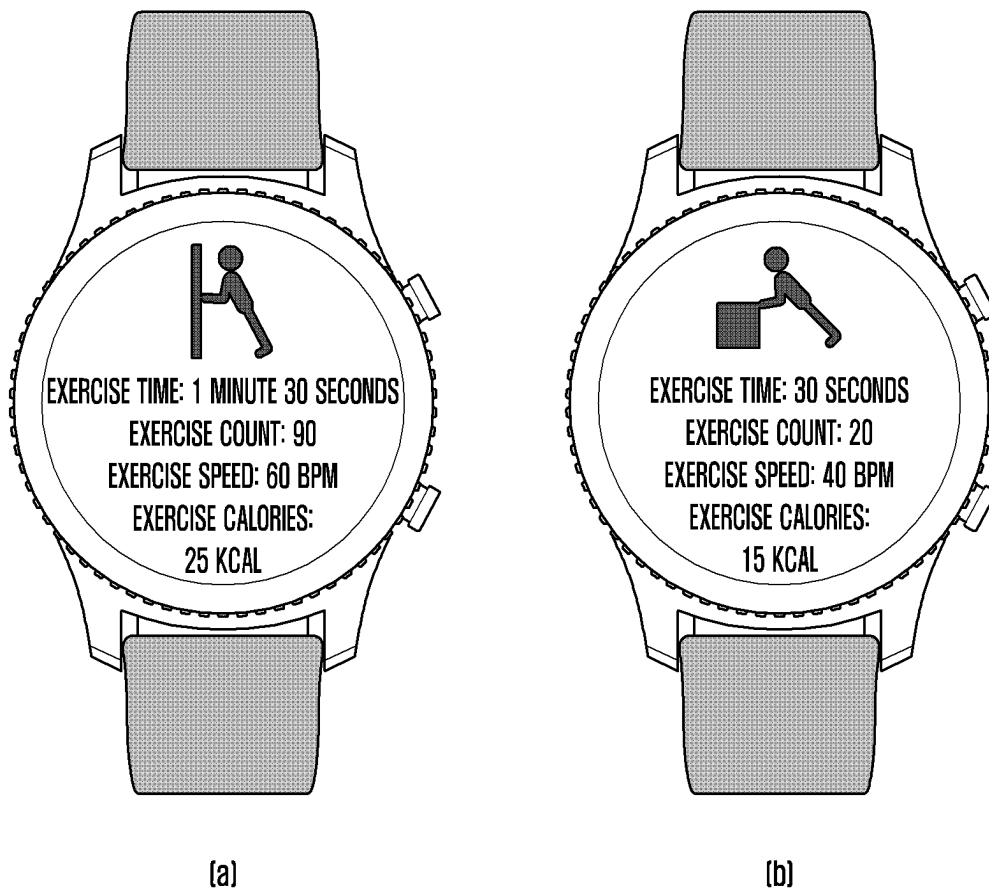
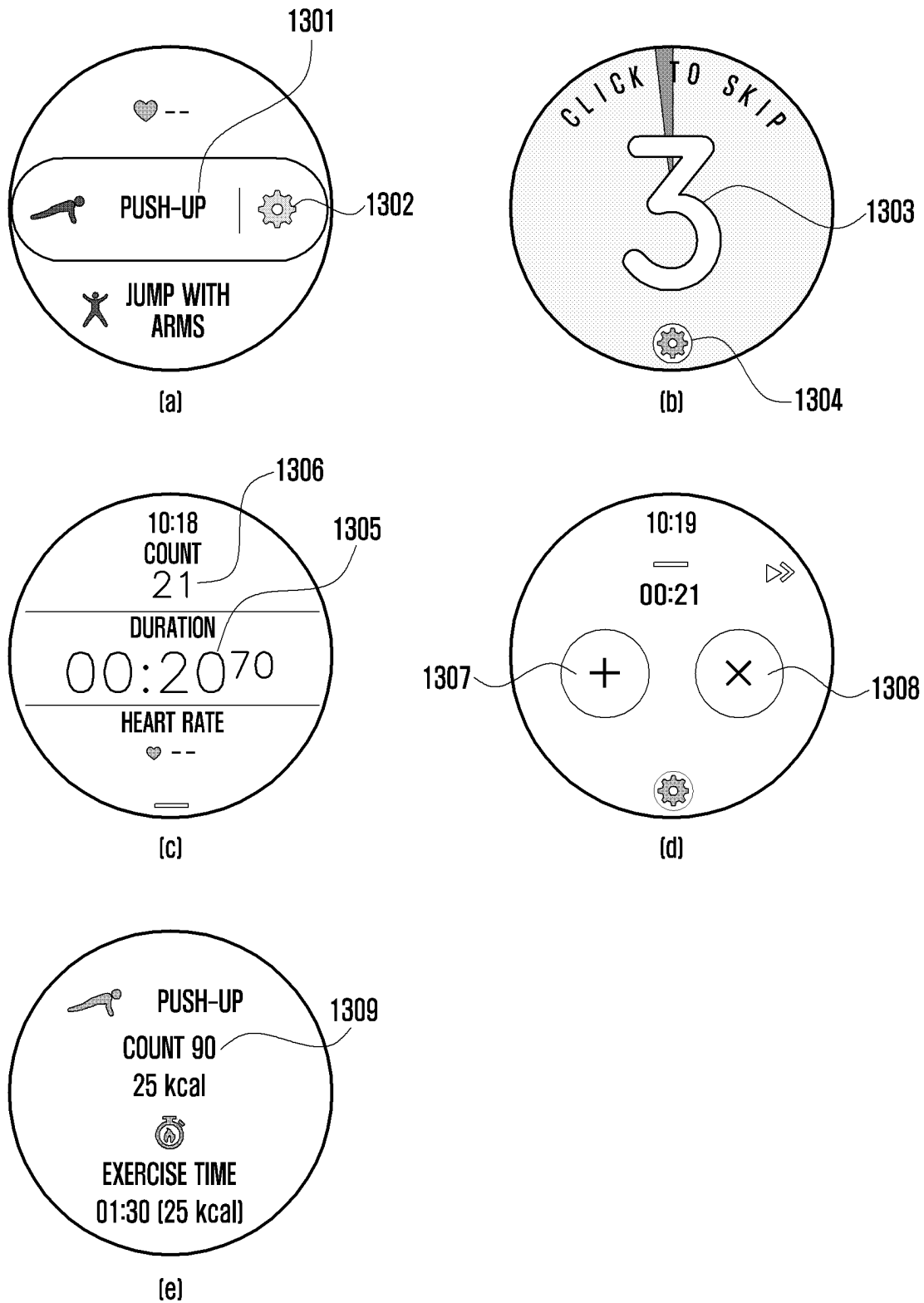


FIG. 13



1

METHOD FOR PROVIDING WORKOUT DATA USING A PLURALITY OF ELECTRONIC DEVICES AND ELECTRONIC DEVICES THEREFOR

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a continuation application, claiming priority under § 365(c), of an International application No. PCT/KR2022/007771, filed on May 31, 2022, which is based on and claims the benefit of a Korean patent application number 10-2021-0070325, filed on May 31, 2021, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

The disclosure relates to a method for providing workout data using a plurality of electronic devices and an electronic device thereof.

BACKGROUND ART

With the development of technology, electronic devices are becoming smaller so that they may be easily carried, and are evolving to perform various functions in various forms of use according to user needs. Various types of wearable devices that may be used by being directly attached to a part of a user's body may be one of the electronic devices.

Recently, attempts have been actively made to obtain various information from the user's body movement by utilizing characteristics of a wearable device directly attached to a part of the user's body and to provide various services based thereon.

The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

DISCLOSURE OF INVENTION

Technical Problem

Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a method and electronic device for providing a user's workout data using a plurality of electronic devices.

Another aspect of the disclosure is to provide a method of obtaining a signal related to a user's movement by interworking and controlling sensors of a plurality of electronic devices and providing workout data according thereto, and an electronic device thereof.

Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

Solution to Problem

In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a display, a communication circuit, and at least one processor

2

electrically connected to the display and the communication circuit, wherein the at least one processor is configured to recognize a user's initial posture based on at least one of motion sensor signals of a first external electronic device and a second external electronic device received through the communication circuit, to control the first external electronic device to receive a first motion sensor signal, and to obtain the user's workout data based on the first motion sensor signal when the initial posture satisfies a designated condition, to control the second external electronic device to receive a second motion sensor signal when the user's movement is not recognized based on the first motion sensor signal and to obtain the user's workout data based on the second motion sensor signal, and to provide the workout data through at least one of the display or the first external electronic device.

In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes a display, a motion sensor, a communication circuit, and at least one processor electrically connected to the display, the motion sensor, and the communication circuit, wherein the at least one processor is configured to recognize a user's initial posture based on at least one of a motion sensor signal received through the motion sensor or a motion sensor signal of an external electronic device received through the communication circuit, to control the motion sensor to receive a first motion sensor signal, and to obtain the user's workout data based on the first motion sensor signal when the initial posture satisfies a designated condition, to control the external electronic device to receive a second motion sensor signal, and to obtain the user's workout data based on the second motion sensor signal when the user's movement is not recognized based on the first motion sensor signal, and to provide the workout data through the display.

In accordance with another aspect of the disclosure, a method of operating an electronic device is provided. The method includes recognizing a user's initial posture based on at least one of motion sensor signals obtained from each of a first external electronic device and a second external electronic device, obtaining, when the initial posture satisfies a designated condition, the user's workout data based on a first motion sensor signal obtained from the first external electronic device, obtaining, when the user's movement is not recognized based on the first motion sensor signal, the user's workout data based on a second motion sensor signal obtained from the second external electronic device, and providing the workout data through at least one of a display of the electronic device or a display of the first external electronic device.

Advantageous Effects of Invention

According to various embodiments, it is possible to more effectively obtain and provide various workout data according to a user's various movements using a plurality of electronic devices.

According to various embodiments, an acquisition performance of workout data according to various situations can be improved by adaptively controlling sensors of a plurality of electronic devices according to the user's various movements to obtain a user's movement-related signal and extracting workout data accordingly.

Further, various effects directly or indirectly identified through this document can be provided.

Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in

conjunction with the annexed drawings, discloses various embodiments of the disclosure.

BRIEF DESCRIPTION OF DRAWINGS

The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following the description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;

FIG. 2 is a block diagram illustrating a plurality of electronic devices according to an embodiment of the disclosure;

FIG. 3 is a conceptual diagram illustrating a method of obtaining a user's workout data using a plurality of electronic devices according to an embodiment of the disclosure;

FIG. 4 is a flowchart illustrating a method of providing a user's workout data according to an embodiment of the disclosure;

FIG. 5 is a flowchart illustrating a method of providing a user's workout data according to an embodiment of the disclosure;

FIG. 6 is a flowchart illustrating an operation of determining initial posture validity according to an embodiment of the disclosure;

FIG. 7 is a flowchart illustrating an operation of obtaining workout data according to an embodiment of the disclosure;

FIG. 8 is a flowchart illustrating an operation of obtaining workout data by controlling a plurality of electronic devices according to an embodiment of the disclosure;

FIGS. 9A, 9B, and 9C are diagrams illustrating an operation of obtaining workout data from various sensor signals according to various embodiments of the disclosure;

FIG. 10 is a diagram illustrating examples of a user interface that provides a guide for obtaining workout data according to an embodiment of the disclosure;

FIG. 11 is a diagram illustrating examples of a user interface that provides workout data according to various exercise items according to an embodiment of the disclosure;

FIG. 12 is a diagram illustrating examples of a user interface that provides workout data exercise items according to an embodiment of the disclosure; and

FIG. 13 is a diagram illustrating examples of a user interface that provides workout data according to an exercise item selection according to an embodiment of the disclosure.

The same reference numerals are used to represent the same elements throughout the drawings.

MODE FOR THE INVENTION

The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

The terms and words used in the following description and claims are not limited to the bibliographical meanings,

but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

It is to be understood that the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a component surface" includes reference to one or more of such surfaces.

FIG. 1 is a block diagram illustrating an electronic device 101 in a network environment 100 according to an embodiment of the disclosure.

Referring to FIG. 1, the electronic device 101 in the network environment 100 may communicate with an electronic device 102 via a first network 198 (e.g., a short-range wireless communication network), or at least one of an electronic device 104 or a server 108 via a second network 199 (e.g., a long-range wireless communication network). According to an embodiment, the electronic device 101 may communicate with the electronic device 104 via the server 108. According to an embodiment, the electronic device 101 may include a processor 120, memory 130, an input module 150, a sound output module 155, a display module 160, an audio module 170, a sensor module 176, an interface 177, a connecting terminal 178, a haptic module 179, a camera module 180, a power management module 188, a battery 189, a communication module 190, a subscriber identification module (SIM) 196, or an antenna module 197. In some embodiments, at least one of the components (e.g., the connecting terminal 178) may be omitted from the electronic device 101, or one or more other components may be added in the electronic device 101. In some embodiments, some of the components (e.g., the sensor module 176, the camera module 180, or the antenna module 197) may be implemented as a single component (e.g., the display module 160).

The processor 120 may execute, for example, software (e.g., a program 140) to control at least one other component (e.g., a hardware or software component) of the electronic device 101 coupled with the processor 120, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor 120 may store a command or data received from another component (e.g., the sensor module 176 or the communication module 190) in volatile memory 132, process the command or the data stored in the volatile memory 132, and store resulting data in non-volatile memory 134. According to an embodiment, the processor 120 may include a main processor 121 (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor 123 (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor 121. For example, when the electronic device 101 includes the main processor 121 and the auxiliary processor 123, the auxiliary processor 123 may be adapted to consume less power than the main processor 121, or to be specific to a specified function. The auxiliary processor 123 may be implemented as separate from, or as part of the main processor 121.

The auxiliary processor 123 may control at least some of functions or states related to at least one component (e.g., the display module 160, the sensor module 176, or the commu-

nication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output

module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication mod-

ule or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an

embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is

to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively,” as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

Various embodiments as set forth herein may be implemented as software (e.g., the program 140) including one or more instructions that are stored in a storage medium (e.g., internal memory 136 or external memory 138) that is readable by a machine (e.g., the electronic device 101). For example, a processor (e.g., the processor 120) of the machine (e.g., the electronic device 101) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer’s server, a server of the application store, or a relay server.

According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

FIG. 2 is a block diagram illustrating a plurality of electronic devices (e.g., a first electronic device 210, a second electronic device 220, and/or a third electronic device 230) according to an embodiment of the disclosure.

Referring to FIG. 2, the first electronic device 210, the second electronic device 220, and/or the third electronic device 230 may include at least some of the components of the electronic device 101 of FIG. 1. According to various embodiments, the first electronic device 210 may be implemented as a wearable electronic device and include, for example, a watch-type wearable device such as a smart watch that has a communication function capable of communicating with the third electronic device 230 and that may be worn on a user’s wrist. According to various embodiments, the second electronic device 220 may be implemented as a wearable electronic device (hereinafter, referred to as an earphone-type wearable device) that may be worn around the user’s ear, such as augmented reality (AR) glasses or an ear bud. For example, the second electronic device 220 may include a headphone, an earphone, an ear bud, smart glasses, and/or AR glasses capable of providing a sound and/or video to the user based on audio data and/or video data received from a sound source electronic device (e.g., the third electronic device 230) through a communication function. Hereinafter, an example in which the first electronic device 210 is implemented into a smart watch and in which the second electronic device 220 is implemented into an ear bud will be described, but various embodiments may not be limited thereto.

According to various embodiments, the third electronic device 230 may be a portable electronic device and/or a mobile electronic device such as a smart phone, a tablet personal computer (PC), a portable multimedia player (PMP), a personal digital assistant (PDA), a laptop PC, and a wearable device.

According to various embodiments, the first electronic device 210 may include a processor 211 (e.g., the processor 120 of FIG. 1), a memory 212 (e.g., the memory 130 of FIG. 1), a communication circuit 213 (e.g., the communication module 190 of FIG. 1), a touch screen 214 (e.g., the display module 160 of FIG. 1), and a motion sensor 217 (e.g., the sensor module 176 of FIG. 1). According to an embodiment, the motion sensor 217 may include an acceleration sensor 218 and/or a gyro sensor 219. According to an embodiment, the first electronic device 210 may further include a biometric sensor 215 for obtaining a user’s biometric information through a skin region in which blood vessels are

11

positioned while maintaining at least a partial contact state with a human body. The components of each of the first electronic device **210**, the second electronic device **220**, and/or the third electronic device **230** illustrated in FIG. 2 are only an example, and some thereof may be omitted or substituted or integrated as one module according to various embodiments. A detailed description of repeated descriptions of the components described with reference to FIG. 1 among the components of each of the first electronic device **210**, the second electronic device **220**, and/or the third electronic device **230** illustrated in FIG. 2 may be omitted here.

According to various embodiments, the second electronic device **220** may include a processor **221** (e.g., the processor **120** of FIG. 1), a memory **222** (e.g., the memory **130** of FIG. 1), a communication circuit **223** (e.g., the communication module **190** of FIG. 1), a microphone **224** (e.g., the input module **150** of FIG. 1), a speaker **225** (e.g., the sound output module **155** of FIG. 1), and a motion sensor **227** (e.g., the sensor module **176** of FIG. 1). According to an embodiment, the second electronic device **220** may further include a touch sensor (not illustrated) configured to detect a touch. According to an embodiment, the motion sensor **227** may include an acceleration sensor **228** and/or a gyro sensor **229**. According to an embodiment, the second electronic device **220** may further include a barometric pressure sensor **226** (e.g., the sensor module **176** of FIG. 1) for identifying a height from the ground according to barometric pressure measurement. According to an embodiment, the second electronic device **220** may obtain a sound through the microphone **224** to convert the sound into an electrical signal by the processor **221** or may output an electrical signal processed by the processor **221** based on audio data to the outside through the speaker **225**. The components illustrated in FIG. 2 are only an example, and some thereof may be omitted or substituted or integrated as one module according to various embodiments. A detailed description of repeated descriptions of the components described with reference to FIG. 1 among the components described with reference to FIG. 1 among the components illustrated in FIG. 2 may be omitted here.

According to various embodiments, the third electronic device **230** may include a processor **231** (e.g., the processor **120** of FIG. 1), a memory **232** (e.g., the memory **130** of FIG. 1), a communication circuit **233** (e.g., the communication module **190** of FIG. 1), and a touch screen **234** (e.g., the display module **160** of FIG. 1). The components illustrated in FIG. 2 are only an example, and some thereof may be omitted or substituted or integrated as one module according to various embodiments. A detailed description of repeated descriptions of the components described with reference to FIG. 1 among the components illustrated in FIG. 2 may be omitted here.

According to various embodiments, the processor **211**, **221**, or **231** may execute, for example, software (e.g., program), and control at least one other component (e.g., hardware or software component) of the first electronic device **210**, the second electronic device **220**, or the third electronic device **230** connected to the processor **211**, **221**, or **231**, and perform various data processing or operations.

According to various embodiments, the memory **212**, **222**, or **232** may store various data used by at least one component (e.g., the processor **211**, **221**, or **231**) or the motion sensor **217** or **227** of the first electronic device **210**, the second electronic device **220**, or the third electronic device **230**. The data may include, for example, input data or output data for software (e.g., program) and a command related thereto. The memory **212**, **222**, or **232** may include a volatile memory or a non-volatile memory.

12

According to an embodiment, the memory **212**, **222**, or **232** may be coupled to the processor **211**, **221**, or **231** and/or the communication circuit **213**, **223**, or **233**.

According to an embodiment, as at least a part of data processing or computation, the processor **211**, **221**, or **231** may load a command or data received from another component (e.g., the motion sensor **217** or **227** or the communication circuit **213**, **223** or **233**) into a volatile memory, process the command or data loaded into the volatile memory, and store result data in a non-volatile memory.

According to an embodiment, the communication circuit **213**, **223**, or **233** may support establishment of a communication channel through a communication link (e.g., a first communication link **201**, a second communication link **202**, or a third communication link **203**) between the first electronic device **210**, the second electronic device **220**, or the third electronic device **230**, and/or communication through the established communication channel.

According to an embodiment, the first electronic device **210** may directly connect with the second electronic device **220** through the third communication link **203** with various communication methods such as Bluetooth to transmit and receive a signal. According to another embodiment, the third electronic device **230** may connect with the first electronic device **210** and the second electronic device **220** through the first communication link **201** and the second communication link **202** with various communication methods such as Bluetooth to transmit and receive a signal.

According to an embodiment, the communication circuit **213**, **223**, or **233** may include a wireless communication module (e.g., cellular communication module, short-range wireless communication module, or global navigation satellite system (GNSS) communication module) or a wired communication module.

According to an embodiment, the communication circuit **213**, **223**, or **233** may enable the first electronic device **210** and/or the second electronic device **220** to communicate with the third electronic device **230** through the first communication link **201** or the second communication link **202** (e.g., a short-range radio communication network such as Bluetooth, WiFi direct, UWB, or infrared data association (IrDA)).

According to an embodiment, the communication circuit **213**, **223**, or **233** may enable the first electronic device **210** to communicate with the second electronic device **220** through a third communication link **203** (e.g., a short-range wireless communication network such as Bluetooth, WiFi direct, UWB, or infrared data association (IrDA)).

According to an embodiment, the communication circuit **213**, **223**, or **233** may include an antenna module (not illustrated). The antenna module of the communication circuit **213**, **223** or **233** may transmit or receive a signal and/or power to and from the outside. According to an embodiment, the antenna module of the communication circuit **213**, **223**, or **233** may include one antenna including a radiator formed in a conductive pattern or a conductor formed on a substrate (e.g., PCB).

According to an embodiment, the motion sensor **217** or **227** may include an acceleration sensor **218** or **228** and a gyro sensor **219** or **229**. The acceleration sensor **218** or **228** and/or the gyro sensor **219** or **229** may detect a movement and/or inertia of the first electronic device **210** or the second electronic device **220**. The acceleration sensor **218** or **228** and/or the gyro sensor **219** or **229** may include a circuit (e.g., integrated circuit (IC)) that controls an operation of the acceleration sensor **218** or **228** and/or the gyro sensor **219** or **229**. For example, the circuit (e.g., integrated circuit (IC))

13

that controls the operation of the acceleration sensor **218** or **228** and/or the gyro sensor **219** or **229** may be included in the first electronic device **210** or the second electronic device **220**, and be implemented into the processor **211** or **221**.

According to an embodiment, the processor **211** or **231** of the first electronic device **210** or the third electronic device **230** may calculate workout data based on a motion sensor signal obtained from the acceleration sensor **218** or **228** and/or the gyro sensor **219** or **229** of the motion sensor **217** or **227**.

According to an embodiment, as the user performs exercise, the third electronic device **230** may control the first electronic device **210** and the second electronic device **220** to obtain workout data. In this case, the first electronic device **210** and/or the second electronic device **220** may obtain motion information according to the user's exercise under the control of the third electronic device **230** to transmit the motion information to the third electronic device **230**. In this case, the third electronic device **230** may communicate with the first electronic device **210** and/or the second electronic device **220** using the first communication link **201** and/or the second communication link **202** (e.g., the first network **198** of FIG. 1) including a short-range communication network such as Bluetooth (or BLE), WiFi direct, ultra wide band (UWB), or infrared data association (IrDA).

According to various embodiments, the processor **231** of the third electronic device **230** may control the first electronic device **210** and/or the second electronic device **220** to receive a motion sensor signal from the first electronic device **210** and/or the second electronic device **220** and recognize the user's initial posture based on the received motion sensor signal. The processor **231** may calculate, for example, a motion value and/or a posture value based on a motion sensor signal received from the first electronic device **210** and/or the second electronic device **220**, and compare the motion value and/or the posture value with a threshold value and/or a configuration value to determine whether the initial posture satisfies a designated condition.

According to various embodiments, in order to obtain workout data, the processor **231** of the third electronic device **230** may control the first electronic device **210** to receive a motion sensor signal obtained by the motion sensor **217** of the first electronic device **210** or may control the second electronic device **220** based on the motion sensor signal from the first electronic device **210** to receive a motion sensor signal obtained by the motion sensor **227** of the second electronic device **220**.

According to various embodiments, the first electronic device **210** may detect a movement of the first electronic device **210** through various types of sensors capable of detecting a movement of the first electronic device **210** such as the gyro sensor **219** and/or the acceleration sensor **218** included in the motion sensor **217** to provide a generated sensor value (e.g., motion sensor signal) to the third electronic device **230** through the communication circuit **213**.

According to various embodiments, the second electronic device **220** may detect a movement of the second electronic device **220** through various types of sensors capable of detecting a movement of the second electronic device **220** such as the gyro sensor **229** and/or the acceleration sensor **228** included in the motion sensor **227** to provide a generated sensor value (e.g., motion sensor signal) to the third electronic device **230** through the communication circuit **223**.

According to various embodiments, the processor **231** of the third electronic device **230** may obtain workout data

14

based on a motion sensor signal received from the motion sensor **217** of the first electronic device **210** to recognize the user's motion.

According to various embodiments, when motion recognition is impossible by a motion sensor signal received from the first electronic device **210**, the processor **231** of the third electronic device **230** may control the second electronic device **220** to receive a motion sensor signal from the second electronic device **220** and to obtain workout data based on the received motion sensor signal, thereby performing the user's motion recognition.

According to various embodiments, the processor **231** of the third electronic device **230** may obtain the user's workout data from a motion sensor signal received from the first electronic device **210** and/or a motion sensor signal received from the second electronic device **220**. For example, the processor **231** of the third electronic device **230** may analyze the motion sensor signal to calculate workout data including an exercise time, the number of exercise, an exercise speed, and/or exercise calories.

According to various embodiments, the processor **231** of the third electronic device **230** may provide various information such as a guide for obtaining workout data and/or the obtained user's workout data through the touch screen **234**. According to an embodiment, the processor **231** of the third electronic device **230** may receive a user input through the touch screen **234**.

According to various embodiments, the touch screen **234** of the third electronic device **230** may provide various visual information related to workout data acquisition by the processor **231**. For example, the touch screen **234** may display various visual information generated based on the workout data or a required visual notification in relation to acquisition of workout data.

According to various embodiments, the processor **231** of the third electronic device **230** may load an application for obtaining workout data stored in the memory **232** and perform operations for obtaining workout data in association with the first electronic device **210** and/or the second electronic device **220**.

According to another embodiment, as the user performs exercise, the first electronic device **210** may obtain motion information according to the user's exercise from a motion sensor signal obtained by the motion sensor **217** thereof or may request a motion sensor signal to the second electronic device **220** to obtain motion information from the received motion sensor signal, thereby obtaining the user's workout data. In this case, the first electronic device **210** may communicate directly with the second electronic device **220** using the third communication link **203** (e.g., the first network **198** of FIG. 1) including a short-range communication network such as Bluetooth (or BLE), WiFi direct, ultra wide band (UWB), or infrared data association (IrDA).

According to various embodiments, in order to obtain workout data, the processor **211** of the first electronic device **210** may receive a motion sensor signal through the motion sensor **217** or receive a motion sensor signal from the second electronic device **220**.

According to various embodiments, the motion sensor **217** of the first electronic device **210** may provide a sensor value (e.g., motion signal) generated according to movement detection of the first electronic device **210** to the processor **211** through various types of sensors capable of recognizing a movement of the first electronic device **210** such as the gyro sensor **219** and/or the acceleration sensor **218**.

According to various embodiments, the processor **211** of the first electronic device **210** may recognize the user's

15

motion through a motion sensor signal received from the motion sensor **217** of the first electronic device **210**.

According to various embodiments, when motion recognition is impossible by a motion sensor signal received from the first electronic device **210**, the processor **211** of the first electronic device **210** may request motion recognition to the second electronic device **220** and receive a motion sensor signal from the second electronic device **220**.

According to various embodiments, the motion sensor **227** of the second electronic device **220** may detect a movement of the second electronic device **220** to provide a generated sensor value (e.g., motion signal) to the first electronic device **210** through various types of sensors capable of detecting a movement of the second electronic device **220** such as the gyro sensor **219** and/or the acceleration sensor **218** according to the control of the processor **221**.

According to various embodiments, the processor **211** of the first electronic device **210** may obtain the user's workout data from a motion sensor signal of the first electronic device **210** received from the motion sensor **217** and/or a motion sensor signal received from the second electronic device **220**. For example, the processor **211** of the first electronic device **210** may analyze a motion sensor signal to calculate workout data including an exercise time, the number of exercise, an exercise speed, and/or exercise calories.

According to various embodiments, the processor **211** of the first electronic device **210** may provide various information such as a guide for obtaining workout data and/or the obtained user's workout data through the touch screen **214**. According to an embodiment, the processor **211** of the first electronic device **210** may receive a user input through the touch screen **214**.

According to various embodiments, the touch screen **214** of the first electronic device **210** may provide various visual information related to workout data acquisition by the processor **211**. For example, the touch screen **214** may display various visual information generated based on the workout data or a required visual notification in relation to acquisition of workout data.

According to various embodiments, the processor **211** of the first electronic device **210** may load an application for obtaining workout data stored in the memory **212** and enable to perform operations for obtaining workout data in association with the second electronic device **220**.

FIG. 3 is a conceptual diagram illustrating a method of obtaining a user's workout data using a plurality of electronic devices according to an embodiment of the disclosure.

Referring to FIG. 3, a user may wear the first electronic device (e.g., the first electronic device **210** of FIG. 2) implemented as a watch-type wearable device on a wrist, wear the second electronic device (e.g., the second electronic device **220** of FIG. 2) implemented as an earphone type wearable device on the ear, and perform exercise. In this case, the user may position the third electronic device (e.g., the third electronic device **230** of FIG. 2) in the vicinity of the user to identify information and/or various notifications according to function execution of various apps including an exercise app for obtaining and providing workout data.

Referring to FIG. 3, in order to perform push-up exercise, the user may hold an initial posture and perform push-up exercise. In an embodiment, push-up exercise is described as an example, but the type of exercise is not limited thereto, and may be applied to various types of exercises in which a

16

part of a body is fixed and in which a movement of another body part is regularly repeated, such as a pull-up or an inverted row.

According to an embodiment, the electronic device (e.g., the electronic device **101** of FIG. 1 or the electronic device **230** of FIG. 2) may include a display (e.g., the display module **160** of FIG. 1 or the touch screen **234** of FIG. 2), a communication circuit (e.g., the communication module **190** of FIG. 1 or the communication circuit **233** of FIG. 2), and at least one processor (e.g., the processor **120** of FIG. 1 or the processor **231** of FIG. 2) electrically connected to the display and the communication circuit, and the at least one processor may be configured to recognize the user's initial posture based on at least one of the first external electronic device (e.g., the first electronic device **210** of FIG. 2) or the second external electronic device (e.g., the second electronic device **220** of FIG. 2) received through the communication circuit, to control the first external electronic device to receive a first motion sensor signal when the initial posture satisfies a designated condition, to obtain the user's workout data based on the first motion sensor signal, to control the second external electronic device to receive a second motion sensor signal when the user's movement is not recognized based on the first motion sensor signal, to obtain the user's workout data based on the second motion sensor signal, and to provide the workout data through at least one of the display or the first external electronic device.

According to an embodiment, the first external electronic device and the second external electronic device may be configured to obtain the first motion sensor signal and the second motion sensor signal, respectively according to the user's movement at different positions among the user's body parts.

According to an embodiment, the designated condition of the initial posture is determined based on at least one of motion sensor signals of the first external electronic device and the second external electronic device, but may include at least one of a condition that the user's movement stops for a specified time or a condition that the designated posture value is maintained.

According to an embodiment, when the user's exercise is not recognized based on the first motion sensor signal, the processor may be configured to transmit a trigger signal requesting to obtain the second motion sensor signal to the second external electronic device.

According to an embodiment, the processor may perform coordinate transformation of the first motion sensor signal or the second motion sensor signal, extract a U-axis signal component, and detect a zero crossing point for the U-axis signal.

According to an embodiment, the processor may detect a zero-crossing section between consecutive zero-crossing points, calculate a peak and a valley within the zero-crossing section, and count the zero-crossing section as the number of one exercise when each of the calculated zero-crossing section, the peak, and the valley is within a designated range.

According to an embodiment, when the user's movement is not recognized based on the second motion sensor signal, the processor may be configured to identify whether the first external electronic device and the second external electronic device provide an UWB ranging function, to perform UWB ranging between the first external electronic device and the second external electronic device to obtain distance information, if the first external electronic device and the second external electronic device provide an UWB ranging function, and to obtain the user's workout data based on the distance information.

17

According to an embodiment, when the user's movement is not recognized based on the second motion sensor signal, the processor may be configured to receive a barometric pressure sensor signal through the barometric pressure sensor of the second external electronic device and to obtain the user's workout data based on the barometric pressure sensor signal.

According to an embodiment, the electronic device (e.g., the first electronic device **210** of FIG. 2) includes a display (e.g., the touch screen **214** of FIG. 2), a motion sensor (e.g., the motion sensor **219** of FIG. 2), a communication circuit (e.g., the communication circuit **213** of FIG. 2), and at least one processor (e.g., the processor **211** of FIG. 2) electrically connected to the display, the motion sensor, and the communication circuit, and the at least one processor may be configured to recognize the user's initial posture based on at least one of a motion sensor signal received through the motion sensor or a motion sensor signal of an external electronic device (e.g., the second electronic device **220** of FIG. 2) received through the communication circuit, to control the motion sensor to receive a first motion sensor signal when the initial posture satisfies a designated condition, to obtain the user's movement data based on the first motion sensor signal, to control the external electronic device to receive a second motion sensor signal when the user's movement is not recognized based on the first motion sensor signal, to obtain the user's workout data based on the second motion sensor signal, and to provide the workout data through the display.

According to an embodiment, the motion sensor and the second external electronic device may be configured to obtain the first motion sensor signal and the second motion sensor signal, respectively according to the user's movement at different positions among the user's body parts.

According to an embodiment, the designated condition of the initial posture is determined based on at least one of motion sensor signals of the motion sensor and the external electronic device, but may include at least one of a condition that the user's movement is stopped or a condition that a designated posture value is maintained for a specified time.

According to an embodiment, when the user's exercise is not recognized based on the first motion sensor signal, the processor may be configured to transmit a trigger signal requesting to obtain the second motion sensor signal to the external electronic device.

According to an embodiment, the processor may perform coordinate transformation of the first motion sensor signal or the second motion sensor signal, extract a U-axis signal component, and detect a zero crossing point for the U-axis signal.

According to an embodiment, the processor may detect a zero-crossing section between consecutive zero-crossing points, calculate a peak and a valley within the zero-crossing section, and count the zero-crossing section as the number of one exercise when each of the calculated zero-crossing section, the peak, and the valley is within a designated range.

According to an embodiment, when the user's movement is not recognized based on the second motion sensor signal, the processor may be configured to identify whether the external electronic device provides an UWB ranging function through the communication circuit, to perform UWB ranging with the external electronic device through the communication circuit to obtain distance information when the external electronic device provides UWB ranging, and to obtain the user's workout data based on the distance information.

18

According to an embodiment, when the user's movement is not recognized based on the second motion sensor signal, the processor may be configured to receive a barometric pressure sensor signal through the barometric pressure sensor of the second external electronic device and to obtain the user's workout data based on the barometric pressure sensor signal.

FIG. 4 is a flowchart illustrating a method of obtaining a user's workout data according to an embodiment of the disclosure.

Referring to FIG. 4, the processor (e.g., the processor **211** or processor **231** of FIG. 2) of the first electronic device (e.g., the first electronic device **210** of FIG. 2) or the third electronic device (e.g., the third electronic device **230** of FIG. 2) may calculate workout data based on the motion sensor signal obtained from the acceleration sensor (e.g., the acceleration sensor **218** or **228** of FIG. 2) and/or the gyro sensor (e.g., the gyro sensor **219** or **229** of FIG. 2) of the motion sensor (e.g., the motion sensor **217** or **227** of FIG. 2) of the first electronic device **210** or the second electronic device (e.g., the second electronic device **220** of FIG. 2).

According to an embodiment, the processor **211** or **231** of the first electronic device **210** or the third electronic device **230** may calculate workout data based on a motion sensor signal obtained from the acceleration sensor **218** or **228** and/or the gyro sensor **219** or **229** of the motion sensor **217** or **227**.

According to an embodiment, the first electronic device **210** may directly connect with the second electronic device **220** through the third communication link **203** with various communication methods such as Bluetooth to transmit and receive a signal. According to another embodiment, the third electronic device **230** may connect with the first electronic device **210** and the second electronic device **220** to transmit and receive a signal through the first communication link **201** and the second communication link **202** with various communication methods such as Bluetooth.

According to various embodiments, in operation **401**, the processor **211** or **231** may recognize the validity of an initial posture for starting exercise based on a motion sensor signal obtained from the motion sensor **217** or **227**.

According to an embodiment, a condition of a valid initial posture may include a state in which a user stops a movement for a predetermined time and takes an initial setup posture for starting exercise.

According to an embodiment, the processor **211** or **231** may determine the validity of the initial posture based on whether a movement is stopped for a predetermined time and/or a designated posture value is maintained. For example, in the case of push-up exercise, a posture facing the ground with a hand on the floor and a body facing down may be taken as an initial posture, the user may identify a posture value measured by placing the hand on the floor in a motion stop state based on a motion sensor signal from the first electronic device **210** worn on a wrist, and identify a posture value measured by viewing the ground in a motion stop state based on a motion sensor signal from the second electronic device **220** worn on the ear.

According to an embodiment, the processor **211** or **231** may recognize the validity of the initial posture based on motion sensor signals from the motion sensor **217** of the first electronic device **210** and the motion sensor **227** of the second electronic device **220**.

According to an embodiment, the first electronic device **210** or the third electronic device **230** may receive an input of an execution command of an exercise app through the touch screen (e.g., the touch screen **214** or **234** of FIG. 2) or

may receive an input of an exercise start command, and accordingly, may start operation **401** for obtaining workout data. According to an embodiment, the first electronic device **210** or the third electronic device **230** may receive an input for selecting one of exercise items provided on the executed exercise app through the touch screen **214** or **234**.

According to an embodiment, in a state in which there is no significant movement based on the motion sensor signal obtained from the sensor module **217** or **227**, when it is identified that the user's heart rate is rising based on a biometric sensor signal of the biometric sensor **215**, the processor **211** or **231** may determine that the user is performing exercise and start an operation for obtaining workout data.

According to an embodiment, in this case, the processor **211** or **231** may determine the validity of the initial posture based on whether a movement is stopped for a predetermined time. For example, when the movement is stopped based on the motion sensor signal obtained from the sensor modules **217** and **227** of the first electronic device **210** and the second electronic device **220** for a predetermined time after an exercise app is executed or an exercise start command is executed, the processor **211** or **231** may determine that the initial posture is valid, and perform an operation for obtaining workout data.

According to an embodiment, as an exercise item is configured or identified, the processor **211** or **231** may connect to the first electronic device **210** and/or the second electronic device **220** with various communication methods such as Bluetooth, and identify the validity of the initial posture.

According to various embodiments, in operation **403**, as the validity of the initial posture is identified, the processor **211** or **231** may start a workout data acquisition operation and receive a first motion sensor signal from the motion sensor **217** of the first electronic device **210**.

According to various embodiments, in operation **405**, the processor **211** or **231** may identify whether a motion is recognized based on the first motion sensor signal. In the case of exercise difficult to measure by the motion sensor **217** of the first electronic device **210**, such as exercise having a small or insignificant movement of the arm according to an exercise item, a motion may not be recognized by the first electronic device **210** worn on the wrist. For example, when a movement is insignificant based on the first motion sensor signal obtained by the sensor module **217** of the first electronic device **210** and the user's heart rate does not rise based on the biometric sensor signal of the biometric sensor **215**, the processor **211** or **231** may determine that there is no user's movement. For example, even if a movement is insignificant based on the first motion sensor signal obtained by the sensor module **217** of the first electronic device **210**, when the user's heart rate rises based on the biometric sensor signal of the biometric sensor **215**, the processor **211** or **231** may determine that there is the user's movement.

According to an embodiment, when an exercise item is selected, if the corresponding exercise item is an exercise item difficult to be recognized by the first electronic device **210** worn on the wrist, the processor **211** or **231** may drive the biometric sensor **215** to identify whether there is a user's movement. For example, in the case of an exercise item difficult to be recognized by the first electronic device **210**, the processor **211** or **231** may identify whether the user's exercise is actually being performed based on a sensor signal of the biometric sensor **215**.

According to various embodiments, when a motion is not recognized based on the first motion sensor signal, the processor **211** or **231** may enable the second electronic device **220** to drive the motion sensor **227** to receive an obtained second motion sensor signal in operation **407**.

According to an embodiment, the processor **211** or **231** may control the second electronic device **220** to drive the motion sensor **227** to obtain the second motion sensor signal, and accordingly the processor **221** of the second electronic device **220** may drive the motion sensor **227** to obtain a second motion sensor signal, and transmit the second motion sensor signal to the first electronic device **210** or the third electronic device **230**.

According to various embodiments, in operation **409**, the processor **211** or **231** may obtain workout data based on the received first motion sensor signal or second motion sensor signal.

According to an embodiment, in the case of exercise in which an arm movement is large according to an exercise item, a motion may be recognized by the first electronic device **210** worn on the wrist. For example, when a motion is recognized by performing various exercise items in which an arm is moved by the first electronic device **210** worn on the wrist, the motion may be recognized by the first electronic device **210**, and accordingly workout data may be obtained based on the first motion sensor signal.

According to an embodiment, in the case of exercise in which a movement of the arm is small or insignificant according to an exercise item, a motion may not be recognized by the first electronic device **210** worn on the wrist. For example, when a motion is not recognized due to a fixed movement of an arm part in the first electronic device **210** worn on the wrist, the second electronic device **220** may recognize a motion according to an action of fixing the arm part and moving a torso, and accordingly, workout data may be obtained based on the second motion sensor signal.

According to an embodiment, the processor **211** or **231** may process a first motion sensor signal or a second motion sensor signal to obtain workout data such as the number of exercises, an exercise time, an exercise speed, or exercise calories. For example, the number of exercises may be obtained by counting one repetition period as one time by analyzing a signal change pattern based on the first motion sensor signal or the second motion sensor signal. For example, the exercise time may be calculated from a time in which it is determined that a motion is recognized and exercise is thus started based on the first motion sensor signal or the second motion sensor signal to a time in which it is determined that the exercise is finished because a motion is not recognized for a certain time or more. For example, the exercise speed may be calculated as the total number of exercises for a total exercise time based on the first motion sensor signal or the second motion sensor signal. For example, exercise calories may be calculated with reference to known calories consumed for a corresponding exercise item in consideration of an exercise time and the number of exercises.

FIG. 5 is a flowchart illustrating a method of providing a user's workout data according to an embodiment of the disclosure.

Referring to FIG. 5, the processor (e.g., the processor **211** or the processor **231** of FIG. 2) of the first electronic device (e.g., the first electronic device **210** of FIG. 2) or the third electronic device (e.g., the third electronic device **230** of FIG. 2) may calculate workout data based on a motion sensor signal obtained from the acceleration sensor (e.g., the acceleration sensor **218** or **228** of FIG. 2) and/or the gyro

21

sensor (e.g., the gyro sensor **219** or **229** of FIG. 2) of the sensor module (e.g., the motion sensor **217** or **227** of FIG. 2) of the first electronic device **210** or the second electronic device (e.g., the second electronic device **220** of FIG. 2).

According to an embodiment, when the processor **211** or **231** of the first electronic device **210** or the third electronic device **230** may not calculate workout data based on a motion sensor signal obtained from the sensor module **217** or **227**, the processor **211** or **231** may calculate workout data based on UWB ranging or sensing data of an barometric pressure sensor (e.g., the barometric pressure sensor **226** of FIG. 2).

According to an embodiment, the processor **211** or **231** of the first electronic device **210** or the third electronic device **230** may calculate workout data based on the motion sensor signal obtained from the acceleration sensor **218** or **228** and/or the gyro sensor **219** or **229** of the sensor module **217** or **227**.

According to an embodiment, the first electronic device **210** may directly connect with the second electronic device **220** through the third communication link **203** with various communication methods such as Bluetooth to transmit and receive a signal. According to another embodiment, the third electronic device **230** may connect with the first electronic device **210** and the second electronic device **220** through the first communication link **201** and the second communication link **202** with various communication methods such as Bluetooth to transmit and receive a signal.

According to various embodiments, in operation **501**, the processor **211** or **231** may identify an exercise item for providing workout data. For example, the first electronic device **210** or the third electronic device **230** may receive an input of an exercise item from the user through a touch screen (e.g., the touch screen **214** or **234** of FIG. 2). For example, the exercise item may include an exercise item in which a specific part of the user's body is fixed and another part thereof is repeatedly moved, such as a push-up, a pull-up, or an inverted row. According to an embodiment, when it is identified that the user's heart rate is increasing based on the biometric sensor signal of the biometric sensor **215** in a state in which there is no significant movement based on the motion sensor signal obtained from the sensor module **217** or **227**, the processor **211** or **231** may determine that the user is performing fitness exercise, and identify the exercise item.

According to various embodiments, in operation **503**, the processor **211** or **231** may recognize the validity of an initial posture for starting exercise based on a motion sensor signal obtained from the sensor module **217** or **227**. According to an embodiment, after identifying an exercise item, the processor **211** or **231** may connect to the first electronic device **210** and/or the second electronic device **220** and receive a motion sensor signal with various communication methods such as Bluetooth, thereby identifying the validity of the initial posture.

According to an embodiment, the processor **211** or **231** may determine the validity of the initial posture based on whether a movement is stopped for a predetermined time and whether a designated posture value is maintained. For example, in the case of push-up exercise, a posture facing the ground with a hand on the floor and a body facing down may be configured as an initial posture. In this case, when a posture value measured by placing the hand on the floor in a motion stop state is identified based on the motion sensor signal from the first electronic device **210** worn on the wrist, and when a posture value measured by viewing the ground in a motion stop state is identified based on the motion

22

sensor signal from the second electronic device **220** worn on the ear, the validity of the initial posture may be recognized.

According to various embodiments, in operation **505**, as the validity of the initial posture is identified, the processor **211** or **231** may drive the motion sensor **217** of the first electronic device **210** to receive the first motion sensor signal.

According to various embodiments, in operation **507**, the processor **211** or **231** may identify whether a motion is recognized based on the first motion sensor signal. For example, in the case of exercise in which a movement of the arm is small or insignificant according to the exercise item, a motion may not be recognized in the first electronic device **210** worn on the wrist.

According to various embodiments, when the motion is not recognized based on the first motion sensor signal, the processor **211** or **231** may receive a second motion sensor signal obtained by driving the motion sensor **227** of the second electronic device **220** in operation **509**.

According to various embodiments, in operation **511**, the processor **211** or **231** may identify whether a motion is recognized based on the second motion sensor signal received from the motion sensor **227**. For example, when a motion by the first motion sensor signal is not recognized due to a fixed movement of the arm part in the first electronic device **210** worn on the wrist, the motion may be recognized by the second motion sensor signal obtained from the second electronic device **220** according to an action of moving a torso.

According to various embodiments, when the motion is not recognized based on the second motion sensor signal, the processor **211** or **231** may identify whether the first electronic device **210** or the third electronic device **230** may perform an UWB ranging function with the second electronic device **220** or whether a sensor signal may be recognized by the barometric pressure sensor **226** of the second electronic device **220** in operation **513**. For example, when the user moves very slowly or stops exercise because it is hard while performing an exercise item such as a push-up, the motion by the second motion sensor signal may not be recognized.

According to various embodiments, if the processor **211** or **231** may recognize UWB ranging or the barometric pressure sensor **226**, the processor **211** or **231** may perform UWB ranging between the first electronic device **210** or the third electronic device **230** and the second electronic device **220** to obtain information on a change in a distance value or may drive the barometric pressure sensor **226** of the second electronic device **220** to obtain information on a change in a height value in operation **515**.

According to various embodiments, in operation **517**, the processor **211** or **231** may provide workout data such as the number of exercise, an exercise time, an exercise speed, or exercise calories based on a first motion sensor signal, a second motion sensor signal, information on a change in a distance value according to UWB ranging, or information on a change in a height value by the barometric pressure sensor **226**.

According to an embodiment, the processor **211** or **231** may count the number of exercises based on a change point of data obtained by processing the first motion sensor signal or the second motion sensor signal. For example, the processor **211** or **231** may detect a point at which a data value changes from a positive value to a negative value or vice versa based on 0, and count an interval between one change point and the next change point as one exercise period.

23

According to an embodiment, the processor **211** or **231** may count the number of exercises based on change information on a distance value according to UWB ranging or change information on a height value by the barometric pressure sensor **226**. For example, the processor **211** or **231** may detect a point in which a change value changes from a positive value to a negative value or vice versa based on 0, or a point in which a distance value or a height value changes from an increase to a decrease or vice versa and count an interval between one change point and the next change point as one exercise period.

FIG. **6** is a flowchart illustrating an operation of determining initial posture validity according to an embodiment of the disclosure.

Referring to FIG. **6**, the processor (e.g., the processor **211** or **231** of FIG. **2**) of the first electronic device (e.g., the first electronic device **210** of FIG. **2**) or the third electronic device (e.g., the third electronic device **230** of FIG. **2**) may determine the validity of an initial exercise posture based on a motion sensor signal obtained from the sensor module (e.g., the motion sensor **217** or **227** of FIG. **2**) of the first electronic device **210** or the second electronic device (e.g., the second electronic device **220** of FIG. **2**).

According to an embodiment, the first electronic device **210** may directly connect with the second electronic device **220** through the third communication link **203** with various communication methods such as Bluetooth to transmit and receive a signal. According to another embodiment, the third electronic device **230** may connect with the first electronic device **210** and the second electronic device **220** through the first communication link **201** and the second communication link **202** with various communication methods such as Bluetooth to transmit and receive a signal.

According to various embodiments, the processor **211** or **231** may determine the validity of an initial posture based on whether a movement is stopped for a predetermined time and a designated posture value is maintained.

According to various embodiments, in operation **601**, the processor **211** or **231** may receive a first motion sensor signal obtained from the motion sensor **217** and/or a second motion sensor signal obtained from the motion sensor **227**.

According to an embodiment, the first motion sensor signal may be acceleration sensor data and/or gyro sensor data obtained from the acceleration sensor **218** and/or the gyro sensor **219** of the motion sensor **217**. According to an embodiment, the second motion sensor signal may be acceleration sensor data and/or gyro sensor data obtained from the acceleration sensor **228** and/or the gyro sensor **229** of the motion sensor **227**.

According to an embodiment, the processor **211** or **231** may receive a plurality of first motion sensor signals obtained by driving the motion sensor **217** for a specified time and/or a plurality of second motion sensor signals obtained by driving the motion sensor **227**.

According to various embodiments, in operation **603**, the processor **211** or **231** may calculate a first motion value and/or a second motion value based on the first motion sensor signal and/or the second motion sensor signal.

According to an embodiment, the first motion value or the second motion value may be a value indicating a magnitude of a movement calculated based on the first motion sensor signal or the second motion sensor signal, respectively. For example, the first motion value or the second motion value a_{NORM} may be calculated as an acceleration 3-axis absolute value (magnitude), as illustrated in Equation 1.

$$a_{NORM} = \sqrt{x^2 + y^2 + z^2}$$

Equation 1

24

According to an embodiment, in operation **605**, the processor **211** or **231** may determine whether an amount of change in the motion value is smaller than a specified threshold value.

According to an embodiment, the processor **211** or **231** may compare an amount of change in a motion value calculated based on a motion sensor signal received for a predetermined time with a specified threshold value. According to an embodiment, the processor **211** or **231** may compare an amount of change in a first motion value calculated based on a first motion sensor signal received for a predetermined time with a specified threshold value. According to an embodiment, the processor **211** or **231** may compare an amount of change in a second motion value calculated based on a second motion sensor signal received for a predetermined time with a specified threshold value. For example, the amount of change ($\text{diff_sum}(k)$) of the motion value for a predetermined time may be calculated based on Equation 2.

$$\text{diff_sum}(k) = \sum_{i=k-N+1}^k \{a_{NORM}(i) - a_{NORM}(i-1)\} \quad \text{Equation 2}$$

According to an embodiment, a valid initial posture may include a state in which the user stops a movement for a predetermined time and takes an initial setup posture for starting exercise. As described above, when an amount of change in the motion value calculated based on a motion sensor signal received for a predetermined time is smaller than a specified threshold value, it may be determined that a movement is in a stopped state. For example, each of amounts of change in a first motion value or a second motion value indicating a movement of each of the first electronic device **210** worn on the wrist and the second electronic device **220** worn on the ear in a state in which a movement of the user's arm and head is stopped and fixed may be smaller than a specified threshold value.

According to an embodiment, when the amount of change in the motion value is greater than or equal to a specified threshold value, it may be determined that the user is in a moving state and does not take a valid initial posture. According to an embodiment, the processor **211** or **231** may provide a notification for inducing the user to take a valid initial posture through the touch screen (e.g., the touch screen **214** or **234** of FIG. **2**).

According to various embodiments, in operation **607**, the processor **211** or **231** may calculate a first posture value and/or a second posture value based on the first motion sensor signal and/or the second motion sensor signal, in operation **609**, the processor **211** or **231** may identify whether each of the first posture value and/or the second posture value is within a configuration value range, and when the posture value is within a configuration value range, the processor **211** or **231** may determine that the initial posture is valid in operation **611**.

According to an embodiment, the processor **211** or **231** may determine the validity of the initial posture based on whether a movement is stopped for a predetermined time and the configured posture value is maintained. For example, in the case of push-up exercise, a posture facing the ground with a hand on the floor and a body facing down may be taken as an initial posture, and the processor **211** or **231** may identify whether a first posture value measured by placing the hand on the floor in a movement stop state based on a first motion sensor signal from the first electronic

25

device **210** worn on the wrist falls a range within the threshold value from a specified configuration value, and identify whether a second posture value measured by viewing the ground in a movement stop state based on a second motion sensor signal from the second electronic device **220** worn on the ear falls a range within a threshold value from a specified preset value.

FIG. 7 is a flowchart illustrating an operation of obtaining workout data according to an embodiment of the disclosure.

Referring to FIG. 7, the processor (e.g., the processor **211** or **231** of FIG. 2) of the first electronic device (e.g., the first electronic device **210** of FIG. 2) or the third electronic device (e.g., the third electronic device **230** of FIG. 2) may obtain workout data based on a motion sensor signal obtained from the sensor module (e.g., the motion sensor **217** or **227** of FIG. 2) of the first electronic device **210** or the second electronic device (e.g., the second electronic device **220** of FIG. 2).

According to an embodiment, the first electronic device **210** may directly connect with the second electronic device **220** through the third communication link **203** with various communication methods such as Bluetooth to transmit and receive a signal. According to another embodiment, the third electronic device **230** may connect with the first electronic device **210** through the first communication link **201** with various communication methods such as Bluetooth to transmit and receive a signal or may be connected to the second electronic device **220** through the second communication link **202** to transmit and receive a signal.

According to various embodiments, in operation **701**, the processor **211** or **231** may extract a U-axis signal component based on a motion sensor signal obtained by the motion sensor **217** of the first electronic device **210** or a motion sensor signal obtained by the motion sensor **227** of the second electronic device **220**.

According to an embodiment, the motion sensor signal obtained by the motion sensor **217** of the first electronic device **210** or the motion sensor **227** of the second electronic device **220** is data about coordinates of a body frame of the first electronic device **210** or the second electronic device **220** and may be converted into coordinates based on an E-axis, an N-axis, and an U-axis of a navigation frame (east north up (ENU)), and a U-axis signal component may be extracted therefrom. Exercise items performed by the user, such as a push-up, a pull-up, or an inverted row, mainly include items performed while moving up and down based on the ground against gravity, and more accurate movement data may be extracted by using U-axis data that may represent a ground-based vertical movement by converting a body frame coordinate axis of the first electronic device **210** or the second electronic device **220** into a navigation frame coordinate axis.

According to various embodiments, the processor **211** or **231** may detect a zero crossing (ZC) section with respect to the U-axis signal component in operation **703**. According to an embodiment, the processor **211** or **231** may detect a point at which the U-axis signal component value increases from (−) to (+) or decreases from (+) to (−) based on a value of 0, a section between consecutive zero-crossing points, for example, may detect from a previous zero-crossing point to a current zero-crossing point as one zero-crossing section. For example, when the processor **211** or **231** detects a point at which the U-axis signal component value decreases from (+) to (−) as a zero crossing point, the processor **211** or **231** may detect a section from the previous decrease point to the current decrease point as one zero-crossing section.

26

According to various embodiments, the processor **211** or **231** may calculate the duration of a zero crossing (ZC) section detected for the U-axis signal component in operation **705**, and calculate a peak and a valley of the U-axis signal component within the zero crossing section in operation **707**.

According to various embodiments, in operation **709**, the processor **211** or **231** may identify whether a length, a peak, and/or a valley of the calculated zero-crossing section is valid. According to an embodiment, the processor **211** or **231** may identify whether the length of the zero-crossing section is within a range greater than or equal to the specified minimum length and less than or equal to the maximum length, and whether each of a peak and/or a valley of the U-axis signal component exists within a range greater than or equal to a specified valley and less than or equal to a maximum value.

According to various embodiments, when the length, the peak, and/or the valley of the zero-crossing section are valid, the processor **211** or **231** may count the corresponding zero-crossing section as the number of valid one exercise in operation **711**. According to an embodiment, the processor **211** or **231** may obtain various workout data such as the number of exercises, an exercise time, exercise intensity, and/or consumed calories according to the count of the number of valid one exercise according to the calculation of the length, the peak, and/or the valley of the zero-crossing section.

FIG. 8 is a flowchart illustrating an operation of obtaining workout data by controlling a plurality of electronic devices according to an embodiment of the disclosure.

FIGS. 9A, 9B and 9C are diagrams illustrating an operation of obtaining workout data from various sensor signals according to various embodiments of the disclosure.

Referring to FIG. 8, the processor (e.g., the processor **211** or **231** of FIG. 2) of the first electronic device (e.g., the first electronic device **210** of FIG. 2) or the third electronic device (e.g., the third electronic device **230** of FIG. 2) may calculate workout data based on a motion sensor signal obtained from the acceleration sensor (e.g., the acceleration sensor **218** or **228** of FIG. 2) and/or the gyro sensor (e.g., the gyro sensor **219** or **229** of FIG. 2) of the motion sensor (e.g., the motion sensor **217** or **227** of FIG. 2) of the first electronic device **210** or the second electronic device (e.g., the second electronic device **220** of FIG. 2).

According to an embodiment, the processor **211** or **231** of the first electronic device **210** or the third electronic device **230** may calculate workout data based on the motion sensor signal obtained from the acceleration sensor **218** or **228** and/or the gyro sensor **219** or **229** of the motion sensor **217** or **227**.

According to an embodiment, when the processor **211** or **231** of the first electronic device **210** or the third electronic device **230** cannot calculate workout data based on the above-described motion sensor signal, the processor **211** or **231** may calculate workout data based on distance change information according to UWB ranging between the second electronic device **220** and the third electronic device **230** or based on a sensor signal of the barometric pressure sensor (e.g., the barometric pressure sensor **226** of FIG. 2) of the second electronic device **220**.

According to an embodiment, the first electronic device **210** may directly connect with the second electronic device **220** through the third communication link **203** with various communication methods such as Bluetooth to transmit and receive a signal. According to another embodiment, the third electronic device **230** may connect with the first electronic

27

device **210** and the second electronic device **220** through the first communication link **201** and the second communication link **202** with various communication methods such as Bluetooth to transmit and receive a signal.

According to various embodiments, in operation **801**, the processor **211** or **231** may perform exercise recognition based on a first motion sensor signal obtained from the motion sensor **217** of the first electronic device **210**.

According to an embodiment, the processor **211** or **231** may perform coordinate transformation of the first motion sensor signal obtained from the motion sensor **217** of the first electronic device **210** into, for example, a navigation frame, extract a U-axis signal component, detect a zero-crossing section, and count the number of exercises, thereby performing exercise recognition.

According to an embodiment, in operation **803**, the processor **211** or **231** may determine whether exercise recognition is possible based on the first motion sensor signal, and if exercise recognition is not possible, the processor **211** or **231** may transmit a signal triggering acquisition of the second motion sensor signal to the second electronic device **220** by the motion sensor **227** in operation **805**.

Referring to FIG. **9A**, it illustrates an example of a signal processing result of a first motion sensor signal, for example, an acceleration sensor signal obtained by the first electronic device **210** when push-up exercise is performed 10 times. With reference to the drawing, an upper graph representing a three-axis signal of a body frame of the acceleration sensor signal represents an x-axis signal component **901**, a y-axis signal component **902**, and a z-axis signal component **903**, and a lower graph represents an U-axis signal **904** extracted after performing coordinate transformation of the three-axis signal of the upper body frame into a navigation frame. Because it is difficult to detect zero crossing **905** from the U-axis signal of the acceleration sensor signal obtained from the first electronic device **210** worn on the wrist while performing push-up exercise having a characteristic in which the hand is fixed during the exercise, it can be seen that exercise recognition is impossible.

According to various embodiments, the processor **211** or **231** may obtain a second motion sensor signal from the second electronic device **220** in operation **807**, and perform exercise recognition based on the second motion sensor signal in operation **809**.

According to an embodiment, the processor **211** or **231** may perform coordinate transformation of the second motion sensor signal obtained from the motion sensor **227** of the second electronic device **220** into, for example, a navigation frame, extract a U-axis signal component, detect a zero-crossing section, and count the number of exercises, thereby performing exercise recognition.

According to various embodiments, in operation **811**, the processor **211** or **231** may identify whether exercise recognition is possible based on the second motion sensor signal, and if exercise recognition is possible, the processor **211** or **231** may obtain workout data in operation **823**.

Referring to FIG. **9B**, it illustrates an example of a signal processing result of a second motion sensor signal, for example, an acceleration sensor signal obtained by the second electronic device **220** when push-up exercise is performed 10 times. With reference to the drawing, an upper graph representing a three-axis signal of a body frame of an acceleration sensor signal represents an x-axis signal component **911**, a y-axis signal component **912**, and a z-axis signal component **913**, and a lower graph represents an U-axis signal **914** extracted after performing coordinate transformation of the three-axis signal of the upper body

28

frame into a navigation frame. Zero crossing **915** may be detected from a U-axis signal of an acceleration sensor signal obtained from the second electronic device **220** worn on the ear while performing push-up exercise having a characteristic that a body including a head moves up and down during the exercise, and for example, a section between a previous zero-crossing point **916** and a current zero-crossing point **917** may be counted as a one exercise execution section, and in the one exercise execution section, a valley **918** and a peak **919** may be detected as workout data.

According to an embodiment, while the user loses strength, for example, due to fatigue accumulation upon performing exercise, and when an exercise speed slows down, extraction of the U-axis signal based on a motion sensor signal obtained through the motion sensor **217** or **227** and accordingly exercise recognition may become impossible.

According to various embodiments, when the processor **211** or **231** is unable to recognize exercise based on the second motion sensor signal, the processor **211** or **231** may identify whether distance information may be obtained according to UWB ranging between the first electronic device **210** and the second electronic device **220** in operation **813**.

According to an embodiment, when distance information may be obtained according to UWB ranging between the first electronic device **210** and the second electronic device **220**, the processor **211** or **231** may perform UWB ranging between the first electronic device **210** and the second electronic device **220** to obtain UWB distance information in operation **815** and recognize exercise based on a change pattern of the UWB distance in operation **817**. For example, the processor **211** or **231** may count a section between a time point at which the distance changes from a previous decrease to an increase and a time point at which the distance changes from a current decrease to an increase as a one exercise execution section from a change pattern of the UWB distance.

According to an embodiment, when distance information may be obtained according to UWB ranging between the first electronic device **210** and the second electronic device **220**, the processor **211** or **231** may enable to perform UWB ranging between the first electronic device **210** and the second electronic devices **220** to obtain UWB distance information in operation **815**, recognize exercise based on a change pattern of the UWB distance in operation **817**, and obtain workout data based on this in operation **823**. For example, the processor **211** or **231** may extract a time point at which the distance changes from a previous decrease to an increase and a time point at which the distance changes from a current decrease to an increase, and count a section between these time points as one time exercise execution section from the change pattern of the UWB distance.

According to an embodiment, when it is impossible to obtain distance information according to UWB ranging between the first electronic device **210** and the second electronic device **220**, the processor **211** or **231** may enable to obtain barometric pressure sensor data from the barometric pressure sensor **226** of the second electronic device **220** in operation **819**, analyze the barometric pressure sensor data to recognize exercise through a periodic change pattern in operation **821**, and for example, extract a time point in which data changes from a previous decrease to an increase and a time point in which data changes from a current decrease to an increase from the change pattern of the

barometric pressure sensor data, and count a section between these time points as a one exercise execution section in operation 823.

Referring to FIG. 9C, the upper graph illustrates an example of a signal processing result of a second motion sensor signal, for example, an acceleration sensor signal obtained by the second electronic device 220 when push-up exercise is performed 10 times. With reference to the drawing, an upper graph representing a three-axis signal of a body frame of the acceleration sensor signal represents an x-axis signal component 921, an y-axis signal component 922, and a z-axis signal component 923, and a middle graph represents an U-axis signal 924 extracted after performing coordinate transformation of the three-axis signal of the upper body frame into the navigation frame. During exercise, because it is difficult to detect zero crossing 925 from the U-axis signal of the acceleration sensor signal obtained from the second electronic device 220 worn on the ear for various reasons, exercise recognition may become impossible. A lower graph is a graph representing a barometric pressure sensor signal value 931 of the barometric pressure sensor 226 of the second electronic device 220, and it can be seen that the barometric pressure sensor signal 931 represents a periodic signal pattern. The processor 211 or 231 may extract, for example, a time point 932 changing from a previous decrease to an increase and a time point 933 changing from a current decrease to an increase from the change pattern of the barometric pressure sensor signal 931, count a section between these time points as a one exercise execution section, and extract a valley 934 and a peak 935 as workout data.

FIG. 10 is a diagram illustrating examples of a user interface that provides a guide for obtaining workout data according to an embodiment of the disclosure.

According to an embodiment, as in an exercise item in which a torso moves against gravity in a state in which a hand is fixed, such as a push-up, a pull-up, or an inverted row, in the case of an exercise item in which it is difficult to obtain workout data with one device (e.g., the first electronic device 210 of FIG. 2), a guide suggesting to wear together another device (e.g., the second electronic device 220 of FIG. 2) and to start measuring the exercise may be displayed on the touch screen (e.g., the touch screen 214 of FIG. 1) of the first electronic device 210.

Referring to part (a), (b), or (c) of FIG. 10, for example, a push-up, pull-up, or inverted row exercise item may display a guide including an image and/or text for the respective exercise along with a guide phrase suggesting to wear together the second electronic device 220.

FIG. 10 illustrates an example of displaying the guide on the touch screen 214 of the first electronic device 210, but this embodiment is not limited thereto, and for example, a guide may be provided through the touch screen 234 of the third electronic device 230 of FIG. 2.

FIG. 11 is a diagram illustrating examples of a user interface that provides workout data exercise items according to an embodiment of the disclosure.

According to an embodiment, as in an exercise item in which a torso moves against gravity in a state in which a hand is fixed, such as a push-up, a pull-up, or an inverted row, in the case of an exercise item in which it is difficult to obtain workout data with one device (e.g., the first electronic device 210 of FIG. 2), as workout data is obtained by wearing together another device (e.g., the second electronic device 220 of FIG. 2) and starting exercise measurement, workout data may be displayed on a touch screen (e.g., the touch screen 214 of FIG. 1) of the first electronic device 210.

Referring to part (a), (b), or (c) of FIG. 11, for example, a push-up, pull-up, or inverted row exercise item may display workout data including an image and/or a text for the respective exercise. For example, the workout data may include an exercise time, the number of exercises, an exercise speed, or exercise calories.

FIG. 11 illustrates an example of displaying the guide on the touch screen 214 of the first electronic device 210, but this embodiment is not limited thereto and for example, a guide may be provided through the touch screen 234 of the third electronic device 230 of FIG. 2.

FIG. 12 is a diagram illustrating examples of a user interface that provides workout data according to various exercise items according to an embodiment of the disclosure.

According to an embodiment, in the case of exercise in which a torso moves obliquely or horizontally instead of up and down in a state in which a hand is fixed, such as a wall pushup and an incline pushup, unlike the examples of FIG. 11, exercise recognition based on the motion sensor signal may be somewhat unstable. In this case, exercise is recognized by the motion sensor signal, but as described above, workout data may be obtained through measurement of a distance between the first electronic device 210 and the second electronic device 220 through UWB ranging and additional exercise recognition based thereon.

According to an embodiment, as in exercise item in which one body part is fixed and another body part is moved, in the case of exercise items in which it is difficult to obtain workout data with one device (e.g., the first electronic device 210 of FIG. 2), another device (e.g., the second electronic device 220 of FIG. 2) may be worn together, exercise may be recognized and workout data may be obtained through UWB ranging as well as a motion sensor signal, and workout data may be displayed on the touch screen (e.g., the touch screen 214 of FIG. 2) of the first electronic device 210 and/or on the touch screen (e.g., the touch screen 234 of FIG. 2) of the third electronic device 230.

Referring to part (a) or (b) of FIG. 12, for example, a wall push-up or incline push-up exercise item may display workout data including an image and/or a text for the respective exercise. For example, the workout data may include an exercise time, the number of exercises, an exercise speed, or exercise calories.

FIG. 13 is a diagram illustrating examples of a user interface that provides workout data according to an exercise item selection according to an embodiment of the disclosure.

According to an embodiment, a user may select an exercise item through a touch screen (e.g., the touch screen 214 of FIG. 2) of the first electronic device 210 and/or a touch screen (e.g., the touch screen 234 of FIG. 2) of the third electronic device 230.

Referring to part (a) of FIG. 13, by selecting and touching through, for example, toggling an icon 1301 including a diagram and/or a text (e.g., push-up) indicating an exercise item, a corresponding exercise item may be selected and performed. Further, detailed configurations (e.g., exercise angle) for the corresponding exercise item may be adjusted through a configuration button 1302.

Referring to part (b) of FIG. 13, for example, when the exercise item (e.g., push-up) selected in part (a) of FIG. 13 is touched, a countdown icon 1303 may be displayed so as to start the exercise item, and after a countdown length has elapsed, exercise measurement may be started. Further, detailed configurations (e.g., countdown length) related to countdown may be adjusted through the configuration but-

31

ton **1304**. For example, when the countdown icon **1303** is touched, exercise measurement may be immediately started.

Referring to part (c) of FIG. **13**, as exercise measurement is started, exercise information such as an exercise time **1305** and/or the number of exercises **1306** may be provided.

According to various embodiments, in the case of exercise items in which it is difficult to obtain workout data with one device (e.g., the first electronic device **210** of FIG. **2**), another device (e.g., the second electronic device **220** of FIG. **2**) may be worn together, and exercise may be recognized and workout data may be obtained through UWB ranging and/or the barometric pressure sensor **226** of the second electronic device **220** as well as a motion sensor signal.

Referring to part (d) of FIG. **13**, when it is identified that a movement has been stopped for a specified time or more through the first electronic device **210** and the second electronic device **220**, for example, an icon **1307** for selecting another exercise item and/or an icon **1308** for selecting exercise stop may be provided.

Referring to part (e) of FIG. **13**, when the exercise stop icon **1308** is selected in part (d) of FIG. **13**, various exercise data **1309** may be displayed on the touch screen **214** of the first electronic device **210** based on the obtained workout data. For example, the workout data may include an exercise time, the number of exercises, or exercise calories.

While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

The invention claimed is:

1. An electronic device, comprising:

a display;

a communication circuit; and

at least one processor electrically connected to the display and the communication circuit,

wherein the at least one processor is configured to:

recognize a user's initial posture based on a motion sensor signal of a first external electronic device received through the communication circuit,

when the initial posture satisfies a designated condition, control the first external electronic device to transmit, to the electronic device, a first motion sensor signal, and obtain the user's workout data based on the first motion sensor signal,

when the user's movement is not recognized based on the first motion control signal, control a second external electronic device to transmit, to the electronic device, a second motion sensor signal and obtain the user's workout data based on the second motion sensor signal, and

display the workout data on at least one of the display or a display included in the first external electronic device.

2. The electronic device of claim 1, wherein the first external electronic device and the second external electronic device are configured to obtain the first motion sensor signal and the second motion sensor signal, respectively according to the user's movement at different positions among the user's body parts.

3. The electronic device of claim 1, wherein the designated condition of the initial posture is determined based on a motion sensor signal of the first external electronic device, and comprises at least one of a condition that the user's

32

movement stops for a specified time or condition that a designated posture value is maintained.

4. The electronic device of claim 1, wherein the processor is configured to control the second external electronic device to transmit, to the electronic device, the second motion sensor signal, when the user's movement is not recognized based on the first motion sensor signal, in that the processor is configured to transmit a trigger signal, requesting to obtain the second motion sensor signal, to the second external electronic device when the user's exercise is not recognized based on the first motion sensor signal.

5. The electronic device of claim 1, wherein the processor is further configured to:

perform coordinate transformation of the first motion sensor signal or the second motion sensor signal based on an East North Up (ENU) navigation frame, extract a U-axis signal component, and detect a zero crossing point for the U-axis signal component.

6. The electronic device of claim 5, wherein the processor is further configured to:

detect a zero-crossing section between consecutive zero-crossing points, calculate a peak and a valley within the zero-crossing section, and count the zero-crossing section as the number of one exercise when each of the calculated zero crossing section, the peak, and the valley is within a designated range.

7. The electronic device of claim 1, wherein the processor is further configured to:

identify that both the first external electronic device and the second external electronic device provide an ultra-wide band (UWB) ranging function when the user's movement is not recognized based on the second motion sensor signal,

control the first external electronic device and the second external electronic device to perform the UWB ranging between the first external electronic device and the second external electronic device to obtain distance information between the first external electronic device and the second external electronic device, and

obtain the user's workout data based on the distance information between the first external electronic device and the second external electronic device.

8. The electronic device of claim 1, wherein the processor is further configured to:

control the second external electronic device to transmit, to the electronic device, a barometric pressure sensor signal, and

obtain the user's workout data based on the barometric pressure sensor signal, when the user's movement is not recognized based on the second motion sensor signal.

9. An electronic device, comprising:

a display;

a motion sensor;

a communication circuit; and

at least one processor electrically connected to the display, the motion sensor, and the communication circuit,

wherein the at least one processor is configured to:

recognize a user's initial posture based on a motion sensor signal received through the motion sensor, when the initial posture satisfies a designated condition, control the motion sensor to obtain a first motion sensor signal, and obtain the user's workout data based on the first motion sensor signal,

33

when the user's movement is not recognized based on the first motion sensor signal, control an external electronic device to transmit, to the electronic device, a second motion sensor signal, and obtain the user's workout data based on the second motion sensor signal, and

display the workout data on the display.

10. The electronic device of claim 9, wherein the motion sensor and the external electronic device are configured to obtain the first motion sensor signal and the second motion sensor signal, respectively according to the user's movement at different positions among the user's body parts.

11. The electronic device of claim 9, wherein the designated condition of the initial posture is determined based on a motion sensor signal of the motion sensor, and comprises at least one of a condition that the user's movement stops for a specified time or a condition that a designated posture value is maintained.

12. The electronic device of claim 9, wherein the processor is configured to control the external electronic device to transmit, to the electronic device, the second motion sensor signal, and obtain the user's workout data based on the second motion sensor signal, when the user's movement is not recognized based on the first motion sensor signal, in that the processor is further configured to transmit a trigger signal, requesting to obtain the second motion sensor signal, to the external electronic device when the user's exercise is not recognized based on the first motion sensor signal.

13. The electronic device of claim 9, wherein the processor is further configured to:

perform coordinate transformation of the first motion sensor signal or the second motion sensor signal based on an East North Up (ENU) navigation frame, extract a U-axis signal component, and detect a zero crossing point for the U-axis signal component.

14. The electronic device of claim 13, wherein the processor is further configured to:

detect a zero-crossing section between consecutive zero-crossing points, calculate a peak and a valley within the zero-crossing section, and count the zero-crossing section as the number of one exercise when each of the calculated zero crossing section, the peak, and the valley is within a designated range.

15. The electronic device of claim 9, wherein, when the user's movement is not recognized based on the second motion sensor signal, the processor is further configured to:

identify that the external electronic device supports an ultra-wide band (UWB) ranging through the communication circuit,

perform UWB ranging with the external electronic device through the communication circuit to obtain distance information between the electronic device and the external electronic device when the external electronic device provides an UWB ranging function, and

34

obtain user's workout data based on the distance information between the electronic device and the external electronic device.

16. The electronic device of claim 9, wherein, when the user's movement is not recognized based on the second motion sensor signal, the processor is further configured to:

control the external electronic device to transmit, to the electronic device, a barometric pressure sensor signal, and

obtain the user's workout data based on the barometric pressure sensor signal.

17. A method of operating an electronic device, the method comprising:

recognizing a user's initial posture based on a motion sensor signal obtained from a first external electronic device;

obtaining, when the initial posture satisfies a designated condition, the user's workout data based on a first motion sensor signal obtained from the first external electronic device;

obtaining, when the user's movement is not recognized based on the first motion sensor signal, the user's workout data based on a second motion sensor signal obtained from a second external electronic device; and displaying the workout data on at least one of a display included in the electronic device or a display included in the first external electronic device.

18. The method of claim 17, wherein the first external electronic device and the second external electronic device are configured to obtain the first motion sensor signal and the second motion sensor signal, respectively according to the user's movement at different positions among the user's body parts.

19. The method of claim 17, further comprising:

identifying, when the user's movement is not recognized based on the second motion sensor signal, that both the first external electronic device and the second external electronic device provide an ultra-wide band (UWB) ranging; and

when the first external electronic device and the second external electronic device provide a UWB ranging function, controlling the first external electronic device and the second electronic device to perform, the UWB ranging between the first external electronic device and the second external electronic device to obtain distance information between the first external electronic device and the second external electronic device, and obtaining the user's workout data based on the distance information between the first external electronic device and the second external electronic device.

20. The method of claim 17, further comprising:

obtaining, when the user's movement is not recognized based on the second motion sensor signal, the user's workout data based on a barometric pressure sensor signal obtained through a barometric pressure sensor of the second external electronic device.

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